

Renewable Energy Trading Optimization (RETO)

Integrated Project of the Year (IPOY)

Final Report

Project Team (Group 3):

Max Concellon Marimon	Matilda Koffa
Shuai Feng	Emily Larson
Mart van Gassel	Carlos Orduña
Michelangelo Inzerilli	Stijn Plekkenpol

April 2026

Executive Summary

As Europe accelerates its energy transition and renewable energy generation capacity continues to increase, grid operators and asset owners increasingly face the challenges of oversupply. This work, produced in partnership with energy trading software company TokWise, explores economic curtailment: the strategic reduction of power generation to avoid losses and maximize profit. While economic curtailment has traditionally been viewed primarily as a grid limitation, this work considers the potential to use economic curtailment as a proactive solution for solar photovoltaic (PV) portfolio management in Germany.

In this project, we developed a mixed-integer linear programming (MILP) optimization model to simulate trading decisions stepwise across the balancing capacity (aFRR), day-ahead (DA), and intraday (ID) markets. The 3-stage model was run using real market and production data from October 2025 for a solar PV production plan in Germany. The results indicated that participation in the aFRR markets is not possible with the current forecast uncertainty, so a 2-stage model was developed that optimizes across DA and ID segments of the spot market. This simplified model was benchmarked against several strategies: 1) a naive dispatch strategy bidding all energy only in the DA market, 2) a DA and ID optimizer that does not allow curtailment, and 3) a refined version of the 2-stage model which includes a co-located battery energy storage system (BESS).

The results suggest that economic curtailment accounts for a 13% reduction in annual PV production, increasing up to 17% if price volatility and negative price periods increase as expected. The results also highlight the key drivers motivating these curtailment decisions, showing that annual PV plant revenue can increase by over 10 % from this curtailment. This demonstrates a clear economic signal for asset owners to implement such strategies in their dispatching process. Furthermore, the results suggest that co-located BESS can be an effective tool to reduce curtailment, whilst also increasing compensation for PV plants.

However, revenue maximization is not the only objective; we also consider how to maximize overall value from the available resource. The results demonstrate how the introduction of co-located BESS can significantly mitigate the amount of energy production that is wasted through curtailment; even a relatively small 1 MWh storage unit was shown to reduce curtailment by nearly 100 MWh per year.

Finally, the system-wide implications for Germany's future energy systems are considered. Under future scenarios with increased price volatility and negative price periods, PV plant revenue is expected to decrease and the incentives for curtailment will likely increase. Instead of simply adding more capacity, we demonstrate that the intentional and strategic use of economic curtailment could increase the output of Germany's future solar generation by up to 17% without requiring investment in additional plants.

This project supports TokWise by providing a blueprint for optimizing trading strategies to respond to complex and volatile price signals in both current and future market landscapes. While economic curtailment drives revenue for asset and portfolio owners, it is not a sustainable choice for energy systems as a whole. In order to efficiently and effectively utilize renewable energy resources to their greatest potential, economic curtailment must be carefully and strategically deployed alongside additional solutions, such as co-located BESS. This work aims to support the continued development of a more sustainable, resilient, and affordable European energy system.

Acknowledgements

We would like to thank our industry partners at TokWise (Julien Marcenac, Nadia Daneva, and Ivaylo Andreev) for their time and expertise. Thank you for welcoming our team and working with us to explore this topic. We have enjoyed working on such an interesting and meaningful project and we very much hope that the results of our work will provide value to TokWise and its clients.

Our sincere thanks to our academic supervisor, Duarte de Mesquita e Sousa, for his guidance and feedback. Further thanks are extended to Carlos González de Miguel for his diligence in answering questions and supporting our research.

We are grateful to Nina Jablonska, our alumni advisor, for her insightful feedback and suggestions for how to best move the project forward.

Finally, we would like to thank InnoEnergy and the SELECT program leadership for facilitating this experience.

Contents

1	Introduction	1
2	Project Plan	3
2.1	Objectives & Scope	3
2.2	Project Management	3
2.3	Industry Partner: TokWise	4
3	Energy Markets and Curtailment	5
3.1	Economic Curtailment Definition	5
3.2	German Electricity Markets	6
3.2.1	Electricity market description and trading sequences	7
3.2.2	Subsidies and non-market remuneration for renewable energy generation	8
3.3	Economic Curtailment Use Cases	8
3.3.1	Defensive Monetization	9
3.3.2	Offensive Monetization	10
4	Tool Planning and Design	11
4.1	Methodology & Optimization Tool Architecture	11
4.1.1	Decision variables	12
4.1.2	Modelled markets	13
4.2	Data inputs and pre-processing	14
4.2.1	Input generation and market price data	14
4.2.2	EEG subsidy	15
4.2.3	aFRR capacity and EAM price calculation	16

4.2.4	DA, ID, aFRR, and RE generation forecasts	17
4.3	Optimization Constraints Modeling Stages Description	17
4.3.1	Stage 1: Balancing capacity market	18
4.3.2	Stage 2: Day ahead market	20
4.3.3	Stage 3: Intraday and balancing EAM	22
4.4	Key Performance Indicators	25
4.5	Modelling scenarios for tool applications	25
4.5.1	Market Conditions	27
4.5.2	Battery storage sensitivity analysis	29
4.5.3	Dispatch Decision Scenarios	32
5	Results and Discussion	34
5.1	3-Stage Optimization Model Results Across Balancing and Spot En- ergy Markets	34
5.1.1	aFRR Market Commitments	35
5.2	2-Stage Optimization Model Results	36
6	Conclusion	41
A	Project Management	43
B	Obtaining Single Value Estimations for the Optimal aFRR Bidding Price	47
C	Input & Decision Variables for Optimization Model	51
D	3-Stage Optimization Results	55
D.1	First KPI results analysis	55

D.2	Cumulative Revenue: Optimized vs. Benchmark	55
D.3	Revenue Breakdown by Market	57
D.4	Optimization Results Across Three Stages	59
D.5	Decision Comparison: Stage 1 vs. Stage 2 vs. Stage 3	59
D.6	DA and ID Market Analysis	62
D.7	Market Prices & Trading Volumes	62
D.8	Curtailment Analysis	65
D.9	aFRR Market Price Inputs and Threshold	67
D.10	Imbalance Analysis	67
E	Generation Forecast vs Actual — Error Analysis	73
F	Day-Ahead and Intraday Market Prices	75

List of Figures

3.1	Chronological timeline of the wholesale and balancing markets in Germany, for trading of a product to be delivered at time period t	6
3.2	Visualization of Germany’s electricity markets and their respective trading and delivery time windows. Compiled from [14][15][16][17][18][19].	8
4.1	Illustration of the 3-stage market.	12
5.1	aFRR market minimum bid requirement overlaid on top of generation forecast and actual PV output.	36
5.2	Revenue structure and total revenue.	37
5.3	Future scenario revenue breakdown.	38
A.1	Gantt Chart providing information on the project’s work packages and its execution timeline.	44
B.1	Example of a Merit Order List generated from anonymous bid information for a 15 -minute interval.	48
B.2	Example of a the activation probability for aFRR EAM bids in an example 15-minute interval.	49
D.1	Cumulative revenue optimized vs benchmark.	56
D.2	Cumulative revenue optimized vs benchmark.	58
D.3	Optimization results across three stages.	60
D.4	Decision comparison - Stage 1 vs. Stage 2 vs. Stage 3.	61
D.5	Day ahead and Intraday market analysis.	63
D.6	Market prices vs. trading volumes.	64
D.7	Curtailment analysis.	66
D.8	aFRR market price inputs and threshold.	69
D.9	Imbalance comparison — optimized vs benchmark.	71

E.1	Generation forecast vs actual - detailed analysis.	74
F.1	Day ahead and intraday market prices.	76

List of Tables

3.1	Integrated bidding and gate closure timelines for German markets. . .	7
4.1	Input data from TokWise.	15
4.2	Key Performance Indicators.	26
4.3	Battery scenario summary table.	32
5.1	Revenue breakdown across all markets and stages — 5 October 2025. . .	35
5.2	Curtailment summary.	35
5.3	Yearly revenue breakdown across scenarios.	39
5.4	Yearly revenue breakdown across scenarios with future negative prices. . .	39
5.5	Curtailment rates and volumes across modelled scenarios.	40
A.1	Project stakeholders.	45
A.2	Work packages.	46
C.1	Input variables for Stage 1.	52
C.2	Decision and auxiliary variables for Stage 1.	52
C.3	Input variables for Stage 2.	53
C.4	Decision and auxiliary variables for Stage 2.	53
C.5	Input variables for Stage 3.	54
C.6	Decision and auxiliary variables for Stage 3.	54
D.1	Final cumulative revenue.	56
D.2	Energy market revenue and volume statistics.	57
D.3	Summary statistics across Stage 2 and Stage 3.	59
D.4	Price vs. volume statistics.	65

D.5	aFRR market participation and revenue.	68
D.6	Imbalance analysis and comparison statistics.	71
D.7	Imbalance avoidance mechanism analysis.	72
E.1	Generation forecast error statistics.	74
F.1	DA and ID market price statistics.	77

Acronyms

aFRR automatic Frequency Restoration Reserve

BCap aFRR balancing capacity

BESS battery energy storage systems

CET Central European Time

DA day ahead

EAM Energy Activation Market

EEG Renewable Energy Source Act (*Erneuerbare-Energien-Gesetz*)

EPEX European Power Exchange

FCR frequency containment reserve

GW gigawatt

GWh gigawatt hours

ID intraday

KPI key performance indicator

kW kilowatt

kWh kilowatt hours

mFRR manual Frequency Restoration Reserve

MILP mixed integer linear programming

MOL merit order list

MW megawatt

MWh megawatt hour

PV photovoltaic

RE renewable energy

reBAP uniform imbalance price (*regelzonenübergreifender Bilanzkreis-Ausgleichsenergiepreis*)

SoH State of Health

TSO transmission system operator

TW terawatt

TWh terawatt hours

UTC Coordinated Universal Time

VWAP volume-weighted average price

WP work package

Chapter 1

Introduction

The rapid growth of renewable energy production is transforming electricity systems across Europe. Today, abundant solar and wind power generation often exceed the capacity of the grid. This phenomenon leads to curtailment, which occurs when grid operators deliberately reduce the output of renewable generators because the system cannot absorb the available supply. Curtailment has been identified as one of the most pressing challenges in Europe’s future grid [1].

At its core, curtailment reflects the simple imbalance that occurs when the supply of uncontrollable wind or solar power exceeds demand. As the installed capacity of these generators continues to scale, periods of oversupply are becoming more frequent, in a system that lacks the flexibility to shift or store energy, forcing operators to limit production. Increasingly, this dynamic is reinforced not only by physical constraints but also by market signals, especially when electricity prices fall to zero or even turn negative. In 2025 alone, Germany, alongside France and the Netherlands, recorded a combined 539 hours of negative day-ahead prices [1], underscoring how persistent and systemic this imbalance has become. These conditions lead to economic curtailment, which occurs when producers voluntarily reduce output to avoid selling electricity at a loss, as opposed to technical curtailment that is driven by grid limitations. This behaviour is particularly relevant in Europe’s liberalized electricity markets, where generators respond to price signals and optimize their revenues accordingly.

Germany provides a critical case study in this regard. As one of Europe’s most advanced renewable energy markets, it is also among those most affected by curtailment. Along with France, Spain, and the Netherlands, Germany faces some of the highest curtailed volumes in Europe. In 2024, solar curtailment in Germany surged by 97% to 1.5 terawatt hours (TWh) [2] leading to compensation costs of €554 million [3] whilst fossil fuel generation remained at 42% [4], driving up consumer bills and simultaneously leaving more room for fossil fuel generators to supply demand. The trend continued in 2025, with an estimated 9.6 TWh of combined wind and solar generation curtailed, representing nearly 4% of total renewable output [5].

These developments are a direct consequence of the energy transition itself, as renewable capacity expands faster than the infrastructure and flexibility mechanisms needed to support it [3]. As noted in Montel’s European price-sensitive curtailment report, curtailment is best understood as “a structural outcome of the current phase of the energy transition, where renewable capacity has expanded faster than the system’s ability to absorb and shift that energy through demand growth, storage and flexibility” [1]. In this sense, Germany’s experience is particularly instructive, with a mature market already grappling with negative prices and commercial curtailment,

which may foreshadow the trajectory of other European systems [1].

As a transitional challenge, there are several emerging solutions aiming to address curtailment, such as energy storage and grid expansion. However, implementing such solutions demands a deeper understanding of the underlying economic mechanisms driving curtailment, in particular the decision-making processes of individual market participants, which remain largely unexplored. This project addresses that gap by modelling the behaviour of renewable energy portfolio owners engaged in economic curtailment. By capturing the individual trade-offs and incentives faced by such actors, the work aims to shed light on the micro-level rationale behind curtailment decisions. In turn, this work enables a better understanding of how these behaviours scale to the market level, which can support the design of policies and market structures that minimize curtailment and maximize the effective use of renewable energy.

Chapter 2

Project Plan

2.1 Objectives & Scope

The objective of this project is to develop an optimization model for TokWise that explores the impact of strategic economic curtailment under various scenarios. The model is designed for a solar photovoltaic (PV) asset operating within the German electricity market. This scope was chosen in order to be most relevant to TokWise's existing client base and to use readily available data. The model is first tested in a three-stage optimizer across both spot and balancing markets. The model is further applied in a two-stage approach which takes into account practical considerations to simulate a more realistic implementation. Several current- and future-focused scenarios and trading strategies are modelled, including co-located battery storage. This model will add value to TokWise's existing portfolio by providing insight into the decision-making process and market participation behavior of renewable energy portfolio owners.

The project was structured in four phases: market and asset investigation; strategy and algorithm development; tool development; and optimisation and integration. In the first stage, research was conducted to understand the basic mechanisms of multiple electricity markets, the participation of renewable energy assets in those markets, and the opportunities and limitations of economic curtailment. Here, the scope was narrowed down to focus on solar PV generation in the German electricity market. In the second stage, the objectives for the model were defined and a literature review was conducted to examine various model types, resulting in an initial design for the tool. The model was built and tested in the third stage. In the fourth and final stage, the model was improved and applied to various economic curtailment scenarios to obtain results and conclusions.

2.2 Project Management

The project was conducted between September 2025 and April 2026. The work was divided into four phases consisting of nine work packages (WPs). In lieu of a formally defined structure of separate workstreams, team members were assigned to work on each WP based on their skill sets, interest level, and availability. WP leadership was distributed equally among the team members. Additional project management information about WPs, timelines, and stakeholders can be found in Appendix A. The Python code used in this project is available upon request.

2.3 Industry Partner: TokWise

Our industry collaboration partner, TokWise, develops AI-powered energy trading software that helps producers and consumers maximise market returns from their renewable energy assets. They offer energy data management, portfolio forecasting, real-time position management, and commercial optimisation to their clients. Their existing optimisation models are tailored to different European markets, optimising trading decisions across both controllable assets (e.g., batteries) and uncontrollable assets (e.g., wind and solar). Based in Sofia, Bulgaria, the company currently operates in multiple European countries and is looking to expand to new markets. As part of their operations supporting renewable energy generators, an understanding of economic curtailment and the revenue derived from it can support TokWise's optimization agents to determine when it is optimal to engage in such actions. Furthermore, understanding the links of such actions to market-wide events can provide a strategic view of the future relevance of such strategies, which can guide and support future initiatives undertaken by TokWise. Consequently, this project aligns with the industrial needs of TokWise, providing a clearer understanding of how and when economic curtailment can be strategically employed to optimise the trading strategy of a renewable energy generator.

Chapter 3

Energy Markets and Curtailment

In this section, the role of economic curtailment in German energy markets is discussed. We first present a definition of economic curtailment and an overview of German electricity markets and their fundamental operational principles. This is followed by a selection of use cases for economic curtailment, which provides a basic understanding of the driving principles for this type of renewable energy curtailment.

3.1 Economic Curtailment Definition

Economic curtailment refers to the intentional, market-driven decision to partially or completely halt electricity generation from renewable assets, even when those assets are physically capable of producing power [6]. Further discussion with TokWise resulted in the following definition that is used for this project: **“Economic curtailment is the decision to partially or fully stop producing energy from assets in a generation unit’s portfolio due to its own marginalized cost characteristic.”** This definition is supported by scientific literature [7].

Unlike technical curtailment, which is imposed by grid operators to maintain system stability or prevent network congestion, economic curtailment originates from the producer’s own market optimization strategy [6]. It is an active portfolio management decision, based on current or forecasted market conditions, from which the operator compares the expected revenue from producing an additional megawatt hour (MWh) against its marginal cost and the alternative market opportunities available. When the expected marginal gain from generation equals or falls below the marginal loss associated with curtailment, withholding energy becomes the economically rational choice [8].

This phenomenon primarily affects intermittent renewable sources, such as wind and solar PV, whose output is dictated by natural conditions rather than demand patterns. Since these technologies generate during periods that do not always align with consumption, oversupply can lead to low or negative wholesale prices. Under such conditions, curtailment may be the optimal decision from both an operational and financial perspective.

3.2 German Electricity Markets

In this section we present an introduction to the German electricity markets, balancing services, and subsidies which set the conditions that lead to economic curtailment decisions.

The German electricity market is organized in a sequence of three distinct layers, the wholesale spot markets, the power balancing markets (also referred as ancillary services markets), and the imbalance settlement mechanism, which guarantee the stability of the system and the continuous balancing between supply and demand. Figure 3.1 illustrates the chronological timeline of the wholesale and balancing markets in Germany. Elements above the hourly axis reference products of the wholesale spot markets, operated by EPEX or NordPool, whereas elements below the axis reference power and energy trading in the ancillary services markets operated by the relevant transmission system operators (TSOs). Two main actors are referred to in this section. A balance responsible party is a market agent authorized to trade energy in wholesale markets (and by extension subject to imbalance settlements). A balancing service provider is an agent in the power balancing markets [9]. Energy trading through bilateral contracts is another major component of the Germany electricity sector, but due to the gradual transition to liberalized wholesale markets [10] and TokWise’s focus on said trading segments, bilateral energy trading is excluded from this study.

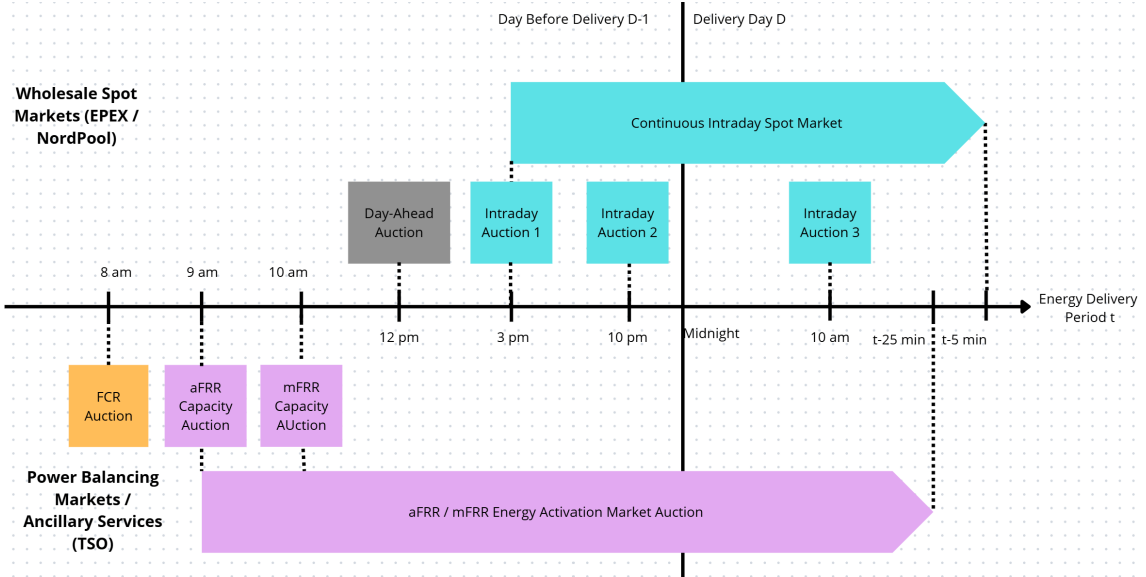


Figure 3.1: Chronological timeline of the wholesale and balancing markets in Germany, for trading of a product to be delivered at time period t .

Market Segment	Product	Gate Closure Time (GCT)	Delivery Period
Balancing Capacity	FCR (Primary Reserve)	D-1 08:00	4-hour
	aFRR (Secondary Reserve)	D-1 09:00	4-hour
	mFRR (Tertiary Reserve)	D-1 10:00	4-hour
Wholesale Spot	Day-Ahead Auction (DAM)	D-1 12:00	15-minute
	Intraday Auction 1 (IDA1)	D-1 15:00	15-minute
	Intraday Auction 2 (IDA2)	D-1 22:00	15-minute
	Intraday Auction 3 (IDA3)	D 10:00	15-minute
	Intraday Continuous	t-5 min	15-minute
Balancing Energy	aFRR / mFRR Energy	t-25 min	15-minute

Table 3.1: Integrated bidding and gate closure timelines for German markets.

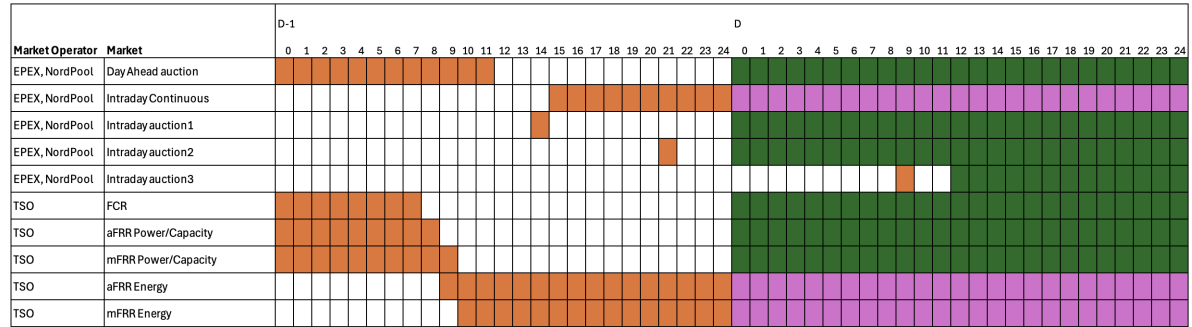
3.2.1 Electricity market description and trading sequences

The main behaviour of markets is centred on the wholesale market, which is the primary layer of energy trading, made up of the day ahead (DA) and intraday (ID) markets [11]. Participants in the market update their positions from year-ahead to five minutes ahead of delivery, thereby forming a merit order list (MOL) through which offers to buy and sell energy are matched, determining how much energy is sold and at what price.

Simultaneously, these merit orders are used by the TSOs in the balancing capacity market to procure frequency containment and restoration reserves to secure grid stability in real-time against deviations [12]. Participation in these markets by balancing service providers can lead to compensation for adjusting energy consumption and production according to the TSO dispatch signals and can provide lucrative revenues in parallel to the main wholesale markets.

Finally, the imbalance settlement mechanism adjusts compensations for the remaining gap between a balance responsible party’s scheduled and actual physical delivery, and is set after energy delivery periods, with the aim of incentivizing proper portfolio management and punishing market agents that do not deliver their promised capacity and lead to system imbalances [13].

The information from Table 3.1 details a chronological sequence of how these markets operate and how decision making is ordered in a cascading fashion, where day D refers to the delivery date of the products, and D-1 refers to the day before D. This structure is shown visually in Figure 3.2, which maps the market interactions, from the start of the DA until the end of the actual delivery day. This figure highlights how the balancing capacity auctions occur mostly in D-1, in contrast to the ID and balancing market that trade parallel to the delivery periods making real time adjustments.



* TSO includes Amprion, TenneT, 50Hertz and TransnetBW

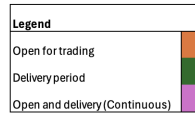


Figure 3.2: Visualization of Germany’s electricity markets and their respective trading and delivery time windows. Compiled from [14][15][16][17][18][19].

3.2.2 Subsidies and non-market remuneration for renewable energy generation

Beyond remuneration through market or imbalance prices, it is also possible to get paid for generating electricity through non-market mechanisms, most prominently through renewable generation subsidies. The main renewable energy subsidies in Germany stem from the Renewable Energy Source Act (*Erneuerbare-Energien-Gesetz*) (EEG) [20], which includes several points such as feed-in tariffs, priority grid access for renewables, and market premium models. Feed-in tariffs ensure that a fixed price is received for each unit of renewable energy produced and delivered to the market [21].

3.3 Economic Curtailment Use Cases

Economic curtailment can generate financial value through two main pathways: by reducing economic losses (defensive monetization) and by creating new revenue opportunities (offensive monetization). Whilst the result of either is a reduction in energy production, their drivers differ significantly. Where defensive monetization is mostly driven by wholesale energy prices and the presence of lack of subsidies, offensive monetization depends on balancing power markets, with imbalance prices affecting both curtailment strategies.

3.3.1 Defensive Monetization

Defensive monetization describes situations where curtailment is implemented to avoid negative returns or high imbalance costs. This can be done by refraining to sell energy in the DA market, or similarly energy can be bought back in the ID market if significant forecast deviations occur.

Negative price avoidance in day ahead market

In the German DA market, defensive monetization commonly arises during extended periods of negative prices [22]. Under such conditions, continuing generation would impose a direct financial loss on renewable producers. This behaviour is strengthened by the negative price rule of EEG [23], through which the market premium is set to zero from the first hour of the negative price period. Consequently, any MWh produced during such hours receives only the negative market price. Producers therefore face a strong incentive to curtail output pre-emptively when forecasts indicate several hours of expected negative prices. Curtailment thus becomes a policy-reinforced form of defensive monetization, where avoided negative revenue is itself an economic gain compared to continued production.

Real-time loss avoidance in intraday market

The intraday market offers a second opportunity to monetize curtailment defensively by adjusting production close to real-time, to reduce contracted volumes or to curtail output directly before delivery. This mechanism is particularly important when forecast errors occur. A producer that initially expected moderate prices may observe a rapid intraday decline caused by unexpected renewable surpluses. Curtailing production avoids selling electricity at extremely low or negative price. However, if a day ahead market bid has already been cleared, considerations must be placed on fulfilling the obligations to produce, since failing to do so will incur paying the imbalance price, which might be more negative than the cleared day ahead market price. To avoid imbalance costs, the sold energy can be re-bought in intraday markets, to ensure the curtailment still adheres to delivery obligations. Because the intraday market closes only five minutes before delivery, it provides an operationally flexible tool for real-time defensive monetization.

3.3.2 Offensive Monetization

Offensive monetization describes situations where the output of flexible assets is deliberately curtailed to participate in more profitable balancing or in ancillary service markets. Such opportunities can be found by participating in balancing power markets, and exploiting favorable imbalance prices.

Shifting from energy markets to balancing capacity markets

Balancing capacity markets allow generators to monetize curtailment by offering flexibility instead of energy. Products such as frequency containment reserve (FCR), automatic Frequency Restoration Reserve (aFRR) and manual Frequency Restoration Reserve (mFRR) are procured through competitive auctions, where providers must commit a minimum of 1 megawatt (MW) of reliable upward or downward capacity, where the latter implies compensation for curtailing production [24]. Participation in such schemes can be further segmented for aFRR and mFRR into capacity markets, which close before the day ahead auction closes, and energy activation markets, open for trading up to 25 minutes before delivery. These markets transform curtailed electricity into a higher-value service, enabling proactive monetization rather than simple loss avoidance.

Curtailment to reduce imbalance exposure and exploit favourable imbalance prices

Another form of offensive monetization arises from strategic imbalance management. In Germany, deviations between scheduled and actual production are settled at the uniform imbalance price (*regelzonenübergreifender Bilanzkreis-Ausgleichsenergiepreis*) (reBAP), which reflects the cost of activating balancing energy [25]. During system-short conditions, reBAP can rise significantly. By curtailing output during periods of uncertainty or fast-changing weather conditions, producers may reduce the risk of being penalized for system-worsening deviations. At the same time, if other assets within the portfolio can offer positive balancing energy, the operator may benefit from high imbalance prices. This approach transforms curtailment into a proactive portfolio strategy aimed at maximizing the value of flexibility rather than minimizing the cost of production. However, difficulties in forecasting reBAP prices increase the difficulty to enact said strategy, which leads to the omission of such an approach due to practical limitations of forecasting methods.

Chapter 4

Tool Planning and Design

4.1 Methodology & Optimization Tool Architecture

The objective of the tool is to determine the financial gain that can be achieved through economic curtailment and, simultaneously, to map out the instances where curtailment would occur to assess its dependence on a range of market conditions. An optimization framework for unit commitment will be employed, in which a given set of inputs are fed into a mixed integer linear programming (MILP) optimization model, which decides the optimal values for the decision variables. The optimal values are therefore the bid positions chosen to maximize the expected returns over all time-steps in a given day, being those the revenues minus the costs of all trades executed across all markets. In this work, a MILP approach is considered due to the flexibility in modelling non-linear market constraints, such as minimum bid volumes and granularity requirements, as well as for its potential to easily integrate with battery energy storage systems (BESS) dispatch modeling.

The optimization approach depends mostly on the assumptions taken on information availability. When trying to find the best global solution, perfect foresight can be assumed, where all information on market prices and unit production is known, and the single best solution can be achieved. While the assumption will generally yield a greater economic gain than realistically possible, it is still a good approximation to highlight expected trading patterns of a rational renewable energy (RE) dispatcher.

When modelling unit commitment of a RE generator through deterministic optimization, the bidding of generation is done according to the timelines of balancing and spot markets, which generally require acting on limited information available before gate closure. To fully model the intricacies of economic curtailment, this sequential nature of electricity markets must be considered. If one single optimization run is taken for each day, the re-adjustment of day-ahead positions through intraday corrections cannot be modelled, thus omitting defensive curtailment. To include both defensive and offensive curtailment in the model outcomes, deterministic sequential optimization can be implemented. As shown in Kraft et al. [26], an optimization run can be done each time a certain market's bidding period closes, determining the optimal bids for all markets and time-periods based on the best available forecasts at the time, and setting the position for the specific market that closes at that time. With the bids for the closed market set as fixed inputs, the optimization can then run again at the next market closure time with updated forecasts, to set the positions for all markets remaining open. Kraft et al. Illustrate such a se-

quential optimization problem in the figure below, where a first optimization run is conducted at 9 AM D-1 to decide the aFRR positions, followed by another stage at 12 PM D-1 to decide the actual day ahead positions (considering aFRR positions as fixed), and finally the last optimization stage runs 25 minutes before every delivery period, to adjust the positions based on the fixed day ahead commitments, and the updated intraday price forecasts available at each time.

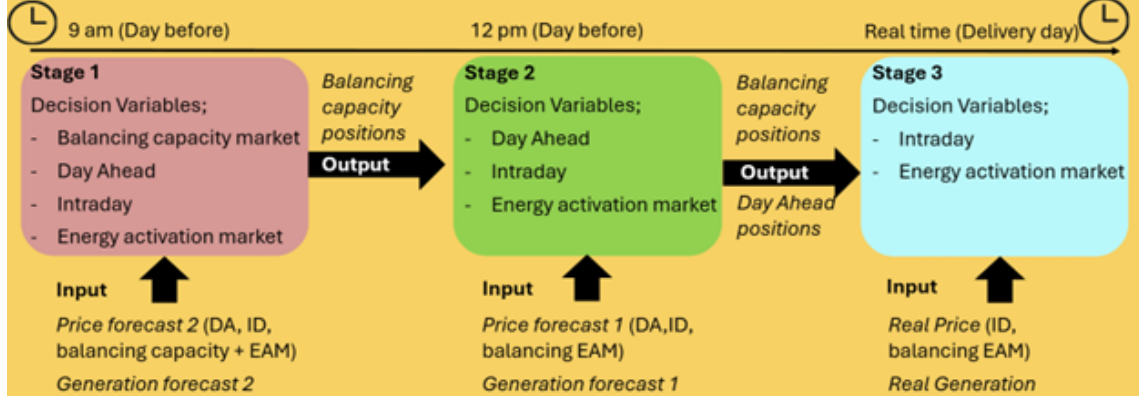


Figure 4.1: Illustration of the 3-stage market.

4.1.1 Decision variables

As a trader dispatching variable RE generation units, the profit is earned based on how much energy is traded at each market, and at which price. These are the decisions a trader must take, namely the volume of energy bid in each market, and the price at which each bid is set. However, the formulation of the problem as a MILP optimization requires that either variable is fixed, and only volume or price can be left as a decision variable.

Considering that the optimization is done with perfect foresight on the prices and output profiles of the variable RE producers, the price level can be fixed as an input variable, leaving the traded volume at each market as the only decision variable. The fixed price associated with each market bid can be derived from market data, based on the type of market remuneration.

Pay-as-cleared markets

It is assumed that the units being dispatched are price takers, meaning they are small enough to not affect the market clearing outcomes. Under this assumption, for pay-as-cleared markets with a uniform price for all participants, which are all markets except continuous intraday and balancing power markets, the remuneration takes the single clearing price for each time-step, as long as the bid price is equal to or below the cleared price. If no uncertainty is assumed, the price can be simply

set as the cleared market price, and only volume is left as a decision variable. The assumption holds true for all deterministic optimization, as a rational market player with perfect information would have no incentive to bid a price above the cleared market price, and would thus always receive said compensation.

Pay-as-bid markets

For pay-as-bid markets such as continuous intraday and m/aFRR capacity markets, the chosen bid price affects the profit together with the volume, and no single clearing price is available. If no uncertainty is modelled, it can be assumed that the optimal price bid with perfect foresight is the maximum capacity price cleared at each time-step, or a more conservative approach could be taken by assuming the average price. In either case, a single price level can be set, leaving only volume as a decision variable for pay-as-bid markets without uncertainty. The optimal bidding price for m/aFRR energy activation markets requires a more nuanced approach, discussed further in the following sections.

4.1.2 Modelled markets

The German energy market has four main segments, namely the capacity reserve, balancing markets, day ahead and intraday markets. For the purpose of this research, the capacity reserve [27] is excluded from the model, given the low relevance for variable RE generators, and the complex bidding and activation dynamics, which for the most part do not influence daily economic curtailment decisions.

Prequalification data [28] [29] for energy players participating in balancing services show that no variable RE generators were prequalified for participation in FCR, or upwards mFRR and aFRR markets. Consequently, these markets will be excluded from the optimization process, as FCR and upwards reserves are not markets accessible to variable RE generators. In Germany, only downwards aFRR and mFRR markets have variable RE producers prequalified for participation, with the mFRR market having the largest prequalified variable RE volume, with 210 MW of wind power. The integration of PV plants into Germany’s secondary balancing reserves (aFRR and mFRR markets) is still underdeveloped, but the first plants prequalified for the aFRR market in late 2025 [30]. Consequently, only the downwards aFRR market will be considered. Both the capacity and energy activation markets for each secondary balancing reserves will be considered.

As the backbone of energy trading, the DA market will be included in the optimization model, together with the ID market. For the latter, four different segments exist, comprised of three auctions and one continuous ID market. To simplify the model, all segments will be condensed into a single auction for each delivery period, as done by Kraft et al. [26] in their sequential optimization modelling for RE unit commitment. This approach greatly simplifies the model, whilst modelling the

dynamics of ID trading to allow for corrections of day ahead positions based on forecast updates, which is a key driver of economic curtailment. To set the price level of the single ID market considered, the volume average weighted price for all ID market segments of a given delivery period will be taken as the set price for any bid.

4.2 Data inputs and pre-processing

The first step of the bid allocation model consists of loading price data for all modelled markets, as well as generation and generation forecast data for the RE assets.

4.2.1 Input generation and market price data

The main elements required for the model are forecasts and actual values of the renewable asset generation, day ahead and intraday market prices and imbalance costs. The input data to the model, provided by TokWise, is described in Table 4.1.

An additional column for `PeriodStartLocal` is included, converting Coordinated Universal Time (UTC) to Central European Time (CET) (for Germany). This will be used as the reference timestamp for the start of each quarter hourly delivery period.

A full year of data for 2025 (January 1st through December 31st) is used for the 2-stage model. For the 3-stage model, we use a one-month range of data for October 2025, chosen based on the availability of the required datasets. The input data initially ranges from 02:15 AM on the 1st of October 2025 until 10:00 AM on the 1st of November 2025.

The column `schedule_mwh` represents the scheduled energy for the DA market, which follows from decisions to allocate the possible capacity to the DA market, as opposed to curtailing energy or allocating the generation to the ID market. Instead, the `fc_mwh_curr` is the most recent forecast for the generation of the assets, which provides a simple generation forecast for the developed model to independently allocate generation to the relevant markets according to the model's own decision logic. For this reason, `fc_mwh_curr` is used as an input for the model, and `schedule_mwh` is excluded. The `da_nomination_mwh`, which is the executed quantity according to the DA price matrix, is dropped, as it also represents scheduling decisions for DA positions that should be made independently by the model developed. The timestamped price data of spot markets, the unit generation forecast and actual generation, are taken as relevant data inputs to the optimization model. The imbalance prices are

used for results evaluation and quantifying final revenues.

Column name	Description	Value type
PeriodEndUTC	Delivery period end (UTC).	datetime
PeriodEndLocal	Delivery period end (CET).	datetime
create_datetime_utc	Snapshot datetime in UTC, forecast and VWAP data is considered to be the most recent as per this timestamp, such that it represents the time when the forecast for the delivery period is available.	datetime
lead_time	Rounded difference in minutes between delivery period start and create datetime, which represents the time difference between a forecast and the actual time period which is being forecasted.	float [min]
fc_mwh_curr	Most recent forecast as per create datetime.	float [MWh]
da_nomination_mwh	Day ahead nomination of the asset which is known as per create datetime.	float [MWh]
schedule_mwh	Day ahead schedule of the asset which is known as per create datetime, the difference to nomination is that nomination takes the day ahead auction price matrix into account and the quantity is the executed quantity.	float
act_mwh	Actual generation which is not known as per create_datetime, it is ex post.	float
LeadTime	lead_time as integer.	int
closest_VWAP	A volume weighted average price of all intraday continuous market trades for this delivery period which happened during the 5 minutes before create datetime, it is known as per create datetime.	float
DA	Day ahead auction clearing price for the delivery period, it is known as per create datetime.	float
ISP	Imbalance price for the delivery period, it is not known as per create datetime, it is ex post.	float
Capacity	Capacity of the assets.	object
fc_mwh_prev	The forecast which was most recent at the previous create datetime.	float
PeriodStartUTC	Delivery period start UTC.	datetime

Table 4.1: Input data from TokWise.

4.2.2 EEG subsidy

Under the current German legislation, solar and wind plants are eligible to receive EEG subsidies, also known as the EEG premium. This compensation is applied to actual generation, defined as the total committed volume in DA and ID markets, and only applied for non-negative price periods [31]. The calculation of the EEG subsidy is ex-post, based on the market premium, as published by the German government [32] and based on the calculated average monthly market values for generation from either solar or wind energy, as published in the open data platform NetzTransparenz [33]. The market premium is then simply calculated as the market premium minus the average market value for each month, with the exception that the subsidy is floored to 0 EUR/MWh during specific periods where the DA price falls below 0 EUR.

For the following model, it is assumed that the EEG premium is known at all decision stages, which would be realistic to the extent to which the RE trader can make an accurate estimation for the monthly subsidy rate. Alternatively, the model could be

modified to use forecasted rates, or more conservative EEG assumptions, as desired by the model user.

4.2.3 aFRR capacity and EAM price calculation

aFRR capacity price data

The aFRR capacity market data can be obtained from the data portal, with data on the marginal price cleared for every 4-hour bidding period [34], and from this data synthetic forecasts are derived. With the aFRR capacity market remunerating offers through a pay as bid system [19], the trader is assumed to bid at the marginal price, as it offers the highest possible compensation, whilst ensuring it is accepted. This is in line with the assumption of partial perfect foresight used for this model, in which the RE trader has a forecast for the marginal price in each price period, and places offers based on said forecasts.

aFRR EAM price data

The EAM is more complex to model, and several steps are taken to obtain a single price per MWh bid at each bidding period, which can then serve as an input to the optimizer. As a member of the PICASSO platform to coordinate aFRR reserves, bids are activated in 4-second blocks, and remunerated based on the total activated energy and the applicable price for each 4-second block. The applicable price is the highest (lowest for negative aFRR) of either the bid price or the local cross border marginal price, depending on each period's needs for energy across borders, so the market is generally operating under pay as bid compensation when capacity is sufficient [35]. Therefore, the strategic bid price will depend on the expected duration of activations, as there is no unique marginal price that guarantees full activation and maximum compensation.

The methodology used to obtain single value estimations for the optimal bidding price and the expected compensation in each 15 minute period follows the activation-simulation approach of Loesch & Rominger (2018) [36] and the activation-duration formulation used in Gärtner et al. (2022) [37]. The raw data needed is found in Regellistung's data center [34] from where data on the anonymous bid lists and quality assured activated volumes can be obtained. The methodology includes five steps, listed here and explained in detail in Appendix B:

1. Build the merit order list from anonymous bids.
2. Determine activation demand each minute.

3. Compute the probability of bid activation for each bid price level.
4. Compute the optimal bidding price.
5. Estimate the expected activation compensation in EUR of revenue per MWh bid.

4.2.4 DA, ID, aFRR, and RE generation forecasts

Synthetic price forecasts are created for ID and aFRR market price data, as well as for a second PV generation forecast, using an autoregressive method to introduce a random error to the real input price data.

Two forecasts are created for market data, forecast 1 which has a lower deviation from real prices, and forecast 2, which represents an earlier and less accurate forecast, and shows a higher deviation from real prices than forecast 1. These synthetic forecasts are used to model forecast uncertainty in the different modelling stages, with growing forecast uncertainty for decision stages with greater lead time, which represents the time difference between a forecast or operational decision, and the actual time being forecasted or the execution time of an action. The solar generation data already has one forecast, so only forecast 2 for 9 AM on the day before delivery (D-1) is created.

Two forecasts for DA prices are created, the first one using data for the previous day's prices, which was suggested by TokWise as a solid strategy to replicate the error and uncertainty expected from such forecasts. Using this forecast, a further synthetic forecasts for 9 AM on the day before delivery is created.

4.3 Optimization Constraints Modeling Stages Description

The model constraints and code structure for each stage is explained below, presenting the modelling architecture for each of the decision stages. The sign convention in the model is consistent with cash flows to the RE asset owner, where a positive sign means revenue paid to the asset owner. Consistently with the convention, energy flows have a positive sign for energy sold by the RE asset owner to the market, and has a negative sign for energy bought by the asset owner in the markets. In this sense, curtailment would have a negative sign, as it subtracts from a positive flow of energy from the asset owner to the market.

Further detail on the input and decision variables of each stage can be found in Appendix C.

4.3.1 Stage 1: Balancing capacity market

The balancing capacity market stage occurs before the bidding period for aFRR (mFRR) closes, which in Germany occurs at 9 AM (10 AM). At this stage the only available inputs to the model are forecasts for market prices and RE generation, which are labelled as forecast 2. This stage allocates part of the forecasted RE generation to the aFRR (mFRR) downwards market. The aFRR downwards market is chosen for solar energy, since it is the only reserve of such type for which PV plants have been prequalified in Germany, with the first such qualification taking place in late 2025 [38].

aFRR bidding upper limits and co-located BESS modelling

A commitment to provide aFRR regulation must always be honored, since failure to provide a committed capacity will lead to disqualification as an aFRR provider. Consequently, upper limits must be placed on the maximum aFRR downwards bid, to avoid forecast deviations from reducing RE plant output below the volume committed in the aFRR downwards market. This is because if generation falls below the committed aFRR volume, the plant owner can no longer provide the level of curtailment promised in the aFRR bid, since the generation is no longer existent, and therefore cannot be curtailed.

This issue is treated widely in the literature, mostly by analysing forecast uncertainty, and applying a reservation margin based on the expected forecast error, which reduces the bid capacity to ensure that the required generation is still reached within expected forecast error scenarios [39]. However, since analyzing forecast errors is beyond the scope of this work, alternative methods found in the literature are considered, such as assigning a fixed reservation margin, as done by Ruter (page 48) [40] when committing only 75% of forecasted generation. In our case, the reservation margin will be estimated from the only prequalified PV plant for negative aFRR provision in Germany [38], which was prequalified for a maximum aFRR volume that is 67% of its peak power, a more conservative estimate.

Whilst the aforementioned method can account for a certain level of uncertainty in RE generation forecasting, the trader must have near full certainty that forecast deviations will not exceed the reservation margin, as this would result in a failure to deliver the promised capacity, and consequently a failure to fulfil market obligations, and subsequently disqualification from said markets.

Whilst the margin can possibly be sufficient to accommodate for forecast deviations between EAM bidding and actual delivery, a greater issue arises from aFRR capacity margin bidding, when a time delta of up to 39 hours must be observed between the time of bidding and actual delivery, for which forecasts can greatly increase in their uncertainty. Consequently, to expand participation to capacity markets, the RE energy plant must have advanced forecasting systems that ensure forecast deviations are consistently below a certain value, chosen as the reservation margin, or co-located BESS must ensure commitments made are honored.

Optimization formula (Stage 1)

This section describes the objective function and constraints for Stage 1 of the model. A detailed explanation of the input and decision variables can be found in Appendix C.

Objective function: The objective function for Stage 1 is shown in Equation 4.1.

$$\begin{aligned} \max \sum_{t \in T} & \left(\frac{DA_t}{4} DA_P_t^{F2} + \frac{ID_t}{4} ID_P_t^{F2} \right. \\ & + \frac{aFRR_N_t}{4} EAM_P_t^{F2} + aFRR_N_t \cdot BCAP_P_t^{F2} \\ & \left. + \frac{DA_t + ID_t}{4} EEG_P_t \right) \end{aligned} \quad (4.1)$$

With decision variables $DA_t \in [0, capacity]$; MW volume expected to be bid in DA market, limited by the installed capacity of the RE plant. $ID_t \in [0, capacity]$; MW volume expected to be bid in the ID market, limited by the installed capacity of the RE plant. $aFRR_N_t \in \{0\} \cup [1, max]$; MW volume bid in aFRR capacity market, where $max = capacity * aFRR_reservation_margin$.

As input data we use the forecast 2 of prices for DA market $DA_P_t^{F2}$, ID market $ID_P_t^{F2}$, EEG subsidy $EEG_P_t^{F2}$, and aFRR EAM $EAM_P_t^{F2}$ and capacity markets $BCAP_P_t^{F2}$, all in EUR/MWh except for capacity market prices in EUR/MW. Since decision variables are in MW, and they represent production for a 15-minute period, their quantities are divided by 4 to obtain the energy delivery in MWh, to be multiplied by the relevant market prices in the corresponding units.

Furthermore, the formulation allows for economic curtailment: the total scheduled volume $DA_t + ID_t$ is constrained to be less than or equal to the forecasted generation. This enables the optimizer to reduce scheduled production if market conditions (e.g., low or negative prices) make it economically optimal. Further economic curtailment through aFRR negative market participation is allowed.

Overall, the objective function determines the optimal allocation of forecasted generation between Day-Ahead and Intraday markets to maximize total expected revenue, including market income and regulatory premiums.

The optimization formula is subject to three constraints: a generation feasibility constraint, an aFRR reservation margin constraint, and an aFRR capacity minimum block constraint.

Generation feasibility constraint: For every delivery period $t \in T$:

$$DA_t + ID_t \leq Generation_t^{F2} \quad (4.2)$$

This constraint ensures that the total scheduled quantity in the Day-Ahead and Intraday markets does not exceed the forecast 2 generation for the corresponding delivery period.

aFRR reservation margin constraint: For every delivery period $t \in T$:

$$aFRR_N_t \leq (DA_t + ID_t) * aFRR_Reservation_Margin \quad (4.3)$$

This constraint limits aFRR bidding within a reservation margin, to ensure promised aFRR curtailment capacity can be honoured even under forecast deviations within the expected range.

aFRR capacity minimum block constraint: For every delivery period $t \in B$, where B are blocks of 16 quarter hourly periods (4 hours), and where $j \in B$ is any other quarter hourly period $j \neq t$ within B :

$$aFRR_N_t = aFRR_N_j \quad (4.4)$$

This constraint limits aFRR bids in the capacity market to be constant for each consecutive 4-hour period of the day, such that all bids in a period have the same value. This is consistent with the 4-hour blocks of the aFRR capacity market.

4.3.2 Stage 2: Day ahead market

The day ahead stage simulates the decision-making process before the DA auction closes at 12 PM. At this time only the balancing capacity market prices and allocated volumes are known, and taken as inputs from Stage 1, and spot price and generation data are only estimated through forecast 1. All other decision variables from Stage

1 can be recalculated according to the most recent forecasts, and the optimized DA volume will be actually bid in this stage.

This section describes the objective function and constraints for Stage 2 of the model. A detailed overview of the input variables, decision and auxiliary variables can be found in Appendix C.

Objective function: The objective function for Stage 2 is shown in Equation 4.5.

$$\begin{aligned} \max \sum_{t \in T} & \left(\frac{DA_t}{4} DA_P_t^{F1} + \frac{ID_t}{4} ID_P_t^{F1} \right. \\ & + \frac{aFRR_N_t}{4} EAM_P_t^{F1} + aFRR_N_fixed_t \cdot BCAP_P_t \\ & \left. + \frac{DA_t + ID_t}{4} EEG_P_t \right) \end{aligned} \quad (4.5)$$

With similar decision variables as in Stage 1, and based on forecast 1 of prices, defined with superscript $F1$. This time aFRR capacity market prices $BCAP_P_t$ and committed volumes $aFRR_N_fixed_t$ are known from Stage 1 results. This formulation also ensures that the generator will not incur an imbalance if forecasts are accurate enough; negative imbalances are eliminated with this constraint, given that all traded energy can be generated, and in case the traded energy is below generation, a positive imbalance can be eliminated through curtailment. However, the possible imbalances from forecast deviations are considered in Stage 3.

Constraints The optimization formula for Stage 2 is subject to a generation feasibility constraint 4.2 and an aFRR reservation margin constraint 4.3, similar to Stage 1. In addition, Stage 2 introduces the aFRR capacity obligation constraint, outlined below.

aFRR capacity obligation constraint: For every delivery period $t \in T$:

$$aFRR_N_fixed_t \leq aFRR_N_t \quad (4.6)$$

This constraint ensures that a volume equal to or greater than the quantity promised in the aFRR capacity market must be bid in the aFRR EAM auction.

Whilst reproducing real market requirements, this constraint introduces a potential infeasibility, since drastic reductions in the forecast below the reservation margin might mean that the capacity promised in Stage 1, based on the first forecast, $aFRR_N_fixed_t$, might not be possibly met if the generation forecast in Stage 2 is lower than the initial forecast times the reservation margin. Under such a condition, the generation feasibility and aFRR reservation margin constraints might render the

model infeasible. This is the same problem faced by RE producers that overestimate their capacity, and can only be solved through a solid knowledge of the maximum expected deviation of forecasts, through installation of co-located BESS to make up for any shortfall of RE production, or by avoiding completely participating in aFRR capacity markets. To enact the last option, the RE generator would simply skip Stage 1 of the optimization problem, and would avoid bidding in the capacity market, resorting only to bid in the EAM, where forecasts are expected to be more reliable.

4.3.3 Stage 3: Intraday and balancing EAM

Stage 3 simulates bidding decisions for the intraday and balancing energy activation markets, when balancing capacity markets and day ahead allocations are already settled, and the remaining decision variables are traded volumes in balancing EAM and intraday markets.

Whilst balancing EAM has quarterly resolution auctions that close 25 minutes before a delivery period, Intraday markets are actually four separate markets, namely the intraday continuous market, with quarterly auctions closing five minutes before each delivery period, and three auctions with a unique closing time, that ranges between 3 PM of the day before delivery, to 10 AM of the delivery day. Trading for different delivery periods is possible in all those markets, which have different price levels and liquidity, which increases the complexity to estimate a single optimal bidding price to trade energy in all intraday markets. For this, a volume weighted average price of trading in all markets, *closest_VWAP* in the input data, can be used as a single trading price to condense trades in all intraday markets to a single simplified intraday price.

Given that Stage 3 is conducted less than 30 minutes before delivery, it is expected that generation and price forecasts are very accurate. To simplify the current model, actual price and generation data are used, under the assumption that forecasts at Stage 3 would be very accurate, nearly equal to real variable values. To further consider forecast uncertainty effects, forecasts of real values could be used in this stage instead of actual values, which would not significantly affect the decisions of economic curtailment, and would simply improve the modelling of real operational curtailment needs due to forecast uncertainty, which is not the focus of this study.

Passive imbalance decision

In Stage 3 of the model, imbalances are left as a decision variable, for the RE producer to decide if it's optimal to incur an imbalance or to buy back oversold

generation. This is what is known as a passive imbalance, which is tolerated in certain European regions [41] [42], as opposed to purposefully bidding energy that was never expected to be generated, which would be considered market manipulation, and therefore is illegal.

In line with the possibility of using passive imbalances to avoid buying back energy in the intraday market after wrong forecasts led to overselling DA energy, this model uses imbalance price estimates as an input to the objective function. Because of the complexity of forecasting such prices, they are considered as fixed values of 100,000 EUR/MWh penalties, such that decisions by the model to incur an imbalance are unrealistic, since it would only be possible for higher intraday buy-back prices, which is not possible. However, whilst TokWise's advise suggests that speculating with passive imbalances is risky and complex, the expected penalization of imbalance prices are left open for model users to test. For risk taking traders that are willing to estimate lower imbalance prices, the model can decide whether it is optimal to incur a passive imbalance instead of buying energy in the Intraday market, which will be subject to the imbalance forecasts used as inputs.

Optimization formula (Stage 3)

This section describes the objective function and constraints for Stage 3 of the model. Information on the input variables, decision and auxiliary variables can be found in Appendix C.

In Stage 3, intraday positions are allowed to go negative, in case the RE trader must buy back energy traded in Stage 2 under unrealistically optimistic forecasts. For this, bounds equal to the generation capacity is set, as it is never needed to buy back more energy than the maximum possible generation capacity.

Stage 3 maximizes the total revenue over all delivery periods $t \in T$ using real price data.

Objective function: The objective function for Stage 3 is shown in Equation 4.7.

$$\begin{aligned}
\max \sum_{t \in T} & \left(\frac{DA_fixed_t}{4} DA_P_t + \frac{ID_t}{4} ID_P_t \right. \\
& + \frac{aFRR_N_t}{4} EAM_P_t + aFRR_N_fixed_t \cdot BCAP_P_t \\
& + \frac{DA_t + ID_t + Imb_pos_t - Imb_neg_t}{4} EEG_P_t \\
& \left. - \frac{Imb_pos_t}{4} * c_pos - \frac{Imb_neg_t}{4} * c_neg \right)
\end{aligned} \tag{4.7}$$

With similar decision variables as in previous stages, expanded to include $imb_pos_t \in [0, capacity]$ and $imb_neg_t \in [0, capacity]$, which are the MW volume of positive or negative imbalance, which is surplus or shortage of energy delivered above the promised DA and ID commitments. The price inputs in this stage lack superscripts because it is assumed the actual prices are equal to the forecast available at this time. Furthermore, it is assumed that imbalances are all penalized at a fixed price (cost) c_{pos} and c_{neg} . At this stage aFRR capacity market prices $BCAP_P_t$, DA market prices DA_P_t and committed volumes $aFRR_N_fixed_t$ and DA_fixed_t are known inputs.

Constraints

The formulation is subject to aFRR reservation margin constraint 4.3, and aFRR capacity obligation constraint 4.6, as outlined in previous stages. Furthermore, the imbalance exclusivity constraint is introduced in Stage 3, and the generation feasibility constraint is modified to allow the plant to incur in an imbalance, as outlined below.

Generation feasibility constraint: For every delivery period $t \in T$:

$$0 \leq (DA_t + ID_t + Imb_pos_t - Imb_neg_t) \leq Generation_t \quad (4.8)$$

This constraint ensures that the total scheduled quantity in the Day-Ahead and Intraday markets does not exceed the generation for the corresponding delivery period, unless imbalances are incurred. This ensures the traded energy is reasonable, such that the RE trader cannot sell energy it is not expected to generate, and cannot sell energy below a negative position.

Imbalance exclusivity constraint: The imbalance exclusivity is ensured through the fixed parameter $BIGM = 100\,000$, the auxiliary binary decision variable $delta_imb_t \in \{1, 0\}$, and the following constraints;

$$Imb_pos_t \leq delta_imb_t * BIGM \quad (4.9)$$

$$Imb_neg_t \leq (1 - delta_imb_t) * BIGM \quad (4.10)$$

This auxiliary constraint ensures only positive or negative imbalances are incurred in each period, preventing large imbalances from cancelling each other out in the same period.

Stage 3 curtailment and imbalance implications

This formulation allows economic curtailment, as the optimizer may choose to schedule less than the forecasted generation if market prices make this economically optimal. No additional physical, ramping, imbalance, or balancing market constraints are included in Stage 3.

This formulation also ensures that the generator will not incur an imbalance if forecasts are accurate enough, and because we assume that Stage 3 forecasts are equal to actual values, imbalances are excluded.

Whilst studying the imbalances caused by forecast uncertainty is beyond the scope of this work, if these are to be modelled Stage 3 can be conducted with forecast data, and any deviations between the DA and ID volumes and real generation will inform the model user of the imbalances caused by a lack of forecast accuracy.

4.4 Key Performance Indicators

Seven key performance indicators (KPIs) were defined for the model: total trading revenue and volume, offensive curtailment revenue and volume, defensive curtailment revenue and volume, EEG subsidy revenue, as described in Table 4.2. These indicators serve to evaluate the magnitude and the extent of economic curtailment. Furthermore, breaking down curtailment figures into offensive and defensive curtailment allows for a better understanding of the drivers behind the wider phenomenon. Finally, including data on the EEG subsidies is relevant to understand the impact of governmental support, which is not linked to the competitive nature of the electricity market.

4.5 Modelling scenarios for tool applications

To properly assess the impact of economic curtailment under current and future conditions, changing market conditions and energy storage solutions are modelled to estimate how such options may affect curtailment. Three subsections are discussed: market conditions, battery storage suitability assessment, and dispatch decision scenarios.

The market conditions subsection (Section 4.5.1) provides a macro level analysis of expected developments in the German electricity market. It examines structural changes such as increasing price volatility, the prevalence and structure of negative

#	KPI	Units	Description
1	Total trading revenue	EUR	Overall collected revenue from all traded positions across all markets.
2	Total curtailment volume	MWh	The share of energy curtailed, i.e. the difference between delivered energy and the maximum capacity possible.
3	Offensive curtailment revenue	EUR	Total revenue raised through downwards balancing markets, (i.e. aFRR and mFRR down for solar and wind, respectively) including both capacity and EAM compensation. It informs of the total revenue that was gained by engaging in offensive curtailment with the RE generation unit.
4	Offensive curtailment volume	MWh	Actual energy curtailed through balancing EAM dispatch, which informs of the real impact of participating in balancing reserve markets.
5	Defensive curtailment savings	EUR	Loss avoided by curtailing energy during negative price periods, compared to the loss incurred if curtailment is not allowed.
6	Defensive curtailment volume	MWh	Actual energy curtailed for defensive purposes, to avoid incurring costs during negative price hours.
7	EEG subsidy revenue	EUR	It informs of the impact of governmental support in the form of renewable energy generation subsidies.

Table 4.2: Key Performance Indicators.

price events, and grid congestion mechanisms. These developments are quantified into scenario-based parameters that can be applied to the historical input data used in the model. By introducing these quantified adjustments, the model can be stress-tested under plausible future market conditions, thereby improving its robustness and relevance for forward-looking applications.

The battery storage suitability assessment subsection (Section 4.5.2) evaluates the potential value of battery storage from the perspective of a PV asset owner. Using the curtailment outputs generated by the model, this section introduces the concepts of curtailment magnitude and temporal concentration. By statistically analysing the volume and clustering of curtailed energy, the report provides an initial screening-level assessment of whether the scale and structure of curtailment justify further consideration of co-located battery investment. This assessment does not constitute a full financial valuation but serves as an early-stage decision-support framework for asset owners seeking to enhance profitability under evolving market conditions.

The dispatch decision scenarios subsection (Section 4.5.3) presents a breakdown of the different dispatch decision scenarios, namely diverging approaches to the scheduling of renewables production. This presents an overview of the optimization dispatch strategies that are modelled, from simple benchmarks to the proposed multi-market strategy outlines in Section 4.3, which serve to visualize the impact of including different markets into the optimization dispatch schedule.

4.5.1 Market Conditions

While it appears that the average wholesale energy price is stabilizing from year to year because of declining gas prices, the average conceals the fact that the gap between high and low prices is widening. In 2025, the prices of electricity in the evenings, during the gas peaks, were 19% higher than the daily average, whereas the prices of electricity in the middle of the day, during the solar peaks, often plunged into negative values [43].

Furthermore, regulations around the EEG market premium payment during negative price periods is to be tightened significantly, such that now subsidies will only apply if the negative price event lasts only one hour instead of three, as was the case for new installations (> 400 kW) commissioned after January 1, 2026 [44]. Such regulatory changes nudge power plant owners to voluntarily cut production during any negative price event to avoid penalties, and are also a strong argument for co-located battery storage to absorb energy generation that would be otherwise lost.

For this work we consider two scenarios that reflect the expected changes in the future German energy mix, namely an increasing amount of negative wholesale prices, and higher price volatility.

Scenario 1: Increasing amount of negative wholesale prices

Negative wholesale prices were recorded in 573 out of a total of 8,760 hours in 2025, compared with 457 out of 8,784 hours in 2024 [45]. This upward trend suggests that negative price events are becoming more frequent as renewable penetration increases. Based on this development, a scenario is constructed in which negative price hours rise to approximately 700 hours in 2026, based on the assumption that the growth pattern of 25% increase yearly continues. Such an increase would significantly strengthen negative price signals during high PV production periods, thereby increasing the likelihood of economic curtailment.

To reflect this change in our developed model, the price time series should be modified by increasing the number of negative price hours within the historical dataset while preserving realistic temporal clustering around midday solar peaks. This adjustment allows the optimization model to reassess bidding and dispatch decisions under more frequent negative price conditions and quantify the resulting impact on curtailment volumes and total revenues. The proposed way to incorporate this is to adapt the dataset used in the standard scenario by adding negative wholesale price hours around the current negative wholesale price hours. This should be done for around 130 hours, which accumulates to around 30 minutes of extra negative price hours per day.

Scenario 2: Higher price volatility in the ID and DA markets

Price volatility is expected to remain elevated as renewable penetration increases, but it manifests differently across markets. In the DA market, volatility mainly appears as a stronger intraday shape (deeper midday depressions and higher evening peaks), while the ID market typically shows higher short-term variability driven by forecast updates and real-time balancing needs. In this report, volatility is represented by the daily price spread, defined as:

$$S_d = \max_{t \in d}(p_t) - \min_{t \in d}(p_t) \quad (4.11)$$

and its annual average:

$$S^- = \frac{1}{d} \sum_{d=1}^D S_d \quad (4.12)$$

computed separately for DA and ID markets. Studies have shown that for every 1% increase in share of renewable energy sources, price volatility changes by around 0.5%. Germany is one of the countries that experiences most changes so 0.6% volatility increase per 1% increase in renewable energy sources share is estimated [46]. The renewable installed capacity has increased by around 11% over the last years, so the estimated price volatility increase is assumed to be 6.6% for both DA and ID markets [47].

For our model, higher volatility can lead to more hours where curtailment becomes optimal (if low-price hours become more negative or closer to zero), and stronger incentives to commit generation to high-price hours via market re-positioning where feasible. To implement this scenario, the model input price series can be transformed day-by-day using:

$$p'_t = p_t^- + \alpha(p_t - p_d^-) \quad (4.13)$$

where p_d is the daily mean price, and $\alpha = 1.066$, the estimated increase in volatility. This preserves the daily average level while widening the intra-day spread, allowing the model to quantify how increased price dispersion changes optimal bids, curtailment frequency, and total revenues.

4.5.2 Battery storage sensitivity analysis

As an upcoming solution to renewable power curtailment, the business case for electrical storage is assessed. The model determines optimal bidding strategies across markets and provides hourly information on delivered energy, curtailment volumes, and prevailing market prices. From these outputs, curtailment can be directly identified as energy that is available but not delivered under given market conditions. This creates a natural starting point to assess whether adding a co-located battery system could recover or preserve economic value that would otherwise be lost.

For TokWise, this assessment helps to identify under which market conditions flexibility solutions such as battery storage are economically relevant and therefore worth further analysis or product development. For asset owners, it provides an initial indication of whether curtailment represents a marginal issue or a structurally significant opportunity that could justify investments in additional flexibility.

From an engineering perspective, the value of battery storage can be evaluated at three different levels. Level 1 considers the system or flexibility value, asking whether the magnitude, timing, and economic relevance of curtailment indicate that storage could play a meaningful role. Level 2 focuses on operational value, where explicit battery dispatch is modeled to estimate achievable revenues from energy and ancillary service markets. Level 3 addresses investment or financial value, where battery costs, revenues, and risks are combined to assess profitability. In practice, these levels are sequential. Level 1 serves as a screening step that determines whether more detailed operational and investment analyses are warranted [48].

In this work, the analysis is deliberately limited to Level 1 system potential value. The objective is not to optimize battery dispatch or to evaluate battery investment profitability, but to determine whether the magnitude, temporal concentration, and economic relevance of curtailment observed in the results suggest that battery storage could create or preserve meaningful value.

Scenario inputs from model outputs

The analysis is based on hourly outputs from the optimization model. These include the available renewable generation G_t , the delivered or committed energy D_t , and the corresponding market settlement price p_t . Curtailment is derived as a post-processing metric and defined as the energy that could have been generated but is not delivered to the grid.

From the hourly curtailment time series, aggregate indicators such as total curtailed energy, curtailment share relative to available generation, and measures of temporal concentration are computed. These indicators are treated as derived inputs to the

battery assessment and are not optimized further within this work.

Calculation logic

The calculation logic translates curtailment outcomes into an indicative measure of flexibility relevance, consistent with a Level 1 system value assessment. Curtailment in hour t , as a result of the model output, is defined as:

$$C_t = \max\{0, G_t - D_t\} \quad (4.14)$$

represents energy that is available, but not delivered due to market outcomes. The total curtailment volume and its share relative to the total available generation provide an initial indication of the scale of the issue. However, scale alone is not sufficient to assess the relevance of battery storage. The temporal concentration of curtailment is also considered, for example, by evaluating how much of total curtailment occurs within a small fraction of hours. High temporal concentration indicates that curtailment occurs in clustered periods, which is more compatible with storage-based flexibility than highly dispersed curtailment. We define periods of negative price in Equation 4.15 and the total negative-price-related curtailment in period T is given by Equation 4.16.

$$v_t = \begin{cases} -p_t, & \text{if } p_t < 0 \\ 0, & \text{otherwise} \end{cases} \quad (4.15)$$

$$V_{neg} = \sum_{t=0}^T C_t * V_t \quad (4.16)$$

This value represents the system-level economic relevance of flexibility, rather than actual battery revenues.

Battery storage is introduced in a highly simplified manner, solely to estimate the upper bound of curtailment that flexibility could absorb. The battery is characterized by a maximum charging power P_{bat} and an energy capacity E_{bat} , without modelling state-of-charge dynamics or round-trip efficiency. The maximum amount of curtailed energy that could be absorbed in hour t is given in Equation 4.17.

$$A_t = \min(C_t, P_{bat}, E_{bat}) \quad (4.17)$$

The corresponding upper-bound reduction in negative-price exposure is approximated as:

$$\Delta R_{neg} = \sum_{t, p_t < 0} A_t * (-p_t) \quad (4.18)$$

This quantity should be interpreted as an indicative upper bound of recoverable economic value, not as realized battery profit.

Determining when flexibility becomes valuable

To determine whether battery storage is worth further consideration, the results are interpreted using a Level 1 screening approach based on the characteristics of curtailment observed in the optimization model outputs. Based on the curtailment share, the relevance of storage solutions can be assessed. The curtailment share is defined as the ratio between total curtailed energy and total available renewable generation over the analysed period:

$$\eta_c = \frac{\sum_t C_t}{\sum_t G_t} \quad (4.19)$$

This indicator measures the overall scale of curtailment in a normalized manner and answers the question of whether curtailment is marginal or structurally significant at the system level.

Empirical evidence from multiple power systems shows that renewable curtailment rates below approximately 1–2% are commonly achieved in systems with sufficient flexibility, including storage and grid reinforcements. Curtailment levels in the range of 1–4% are widely observed and are not necessarily indicative of structural inefficiencies, but they warrant further assessment of temporal concentration and economic relevance. Curtailment exceeding 5% is frequently interpreted in engineering and policy-oriented studies as a sign of structural integration challenges, under which flexibility options such as battery storage become highly relevant [49]. This information is summarized in Table 4.3, and serves to identify whether the curtailment magnitude is sufficiently significant to justify more detailed battery dispatch modelling or investment analysis in subsequent work.

Curtailment rate	Typical interpretation	System implication	Relevance of battery storage
<1-2%	Commonly observed in flexible systems	Indicates sufficient grid and market flexibility	Low
1-4%	Frequently observed in renewable systems	Not necessarily inefficient, but requires analysis of temporal concentration	Moderate
>5%	Often associated with structural integration constraints	Suggests persistent congestion or market imbalance	High

Table 4.3: Battery scenario summary table.

4.5.3 Dispatch Decision Scenarios

To evaluate the effectiveness of the proposed multi-stage optimization, the dispatch of power plants is decided through different approaches, which serve to benchmark and assess the effectiveness of different production scheduling strategies, and their impacts on curtailment levels and revenue. There are five strategies assessed:

- **Day-ahead Only:** All forecasted volume is committed at the time of Day-ahead (DA) market closure, regardless of expected prices or the accuracy of the generation forecast. This model is susceptible to costs incurred during negative-price hours and imbalance fees resulting from forecast deviations.
- **Day-ahead and Intraday Trading (No Curtailment):** Total forecasted volume must be traded on either the DA or Intraday (ID) markets. Based on ID price forecasts at the time of DA closure, the model selects the most optimal market to hedge its position. Additionally, if updated generation forecasts during the operating day indicate a potential imbalance, the position is adjusted on the ID market to mitigate risk.
- **Day-ahead and Intraday Trading with Curtailment:** For each time step, the DA and ID markets are evaluated to ensure they yield positive revenue; electricity is then sold in the highest-priced market. If prices in both markets are negative, generation is curtailed. This scenario also includes the capability to buy back production on the ID market to correct DA forecast errors.
- **Day-ahead and Intraday Trading with Co-located BESS:** Operating under similar constraints as the previous scenario, this model includes a co-located 0.5 MW/1 MWh BESS. The BESS optimizes revenue by shifting generation to high-price periods and minimizes imbalances by charging or dis-

charging to meet initial market commitments, such that the BESS can actively correct any imbalances caused by forecast deviations.

Chapter 5

Results and Discussion

The impact and occurrence of economic curtailment is assessed by modelling the optimal dispatch schedule for a PV power plant, using historical data from Germany in 2025. The model is first tested in its full extent, simulating the optimal dispatch decisions across a three-stage optimizer spanning across spot and balancing markets. Simulating a three-stage decision framework including balancing services presents several challenges, both regarding the processing of data, but also implementation challenges since most PV plants do not have access to such markets, and the lack of accurate forecasts can render such participation impossible, as highlighted in Section 5.1. Consequently, the entire historical data is tested in a two-stage model which assumes no participation in aFRR markets, as is the case for most PV plants in Germany. The results are shown in Section 5.2, which explores the impact of economic curtailment across current and future scenarios (as presented in Section 4.5), simulating the dispatch decisions taken through a range of trading strategies, also including co-located battery storage.

5.1 3-Stage Optimization Model Results Across Balancing and Spot Energy Markets

The 3-stage optimization framework developed in Section 4.3 is tested with real data from October 5th, 2025 of a 1.41 MW PV plant. The production data was increased to 14.11 MW peak, by applying a scale factor of 10 to the original Solar_04 dataset (1.41 MW), which was necessary because the original asset was too small to meet the German aFRR market minimum bid requirement of 1 MW after applying the 67% reservation margin. The relevant KPIs as defined in Chapter 4.4 of this report are evaluated for the aforementioned model simulations.

The simulation results are presented in Table 5.1, which presents KPIs 1 and 7, and Table 5.2, which presents the KPIs relevant to curtailment. The results of the simulation are discussed in further detail in Appendix D.

The model's decision-making framework is strictly divided into three distinct phases. First, aFRR capacity is committed in Stage 1, DA market positions are taken at Stage 2, and ID market hedging occurs at Stage 3. As observed in Table 5.1, negative prices and forecast inaccuracies lead to losses incurred in DA trading and unfavourable imbalances settlements. However, the strong performance in ID auctions and particularly in the aFRR market highlight the importance of a multi-market optimization model, which is able to extract value from multiple revenue sources

Revenue Stream	Amount (EUR)
aFRR BCap — Capacity Revenue	1,200.94
aFRR EAM — Activation Revenue	44.86
Intraday Market	32.67
Day-Ahead Market	-7.06
Imbalance Settlement	-8.64
EEG Subsidy (KPI 7)	8.74
Total Revenue (KPI 1)	1,271.52

Table 5.1: Revenue breakdown across all markets and stages — 5 October 2025.

KPI	Value
Negative DA price periods	70 / 96
Defensive curtailment periods	12 / 96
Offensive curtailment revenue (KPI 3)	€1,245.81
Offensive curtailment volume (KPI 4)	8.09 MWh
Defensive curtailment savings (KPI 5)	€10.07
Defensive curtailment volume (KPI 6)	4.33 MWh
Total curtailment volume (KPI 2)	12.42 MWh

Table 5.2: Curtailment summary.

when DA, traditionally primary sources of revenue, discourages energy production.

5.1.1 aFRR Market Commitments

The balancing markets can provide large benefits for participants, as seen through the large revenues raised in the aFRR market for this simulation. However, the low rate of participation of PV plants is largely due to the strict participation requirements. Unlike in spot markets, where failing to meet commitments leads to a simple imbalance payment, balancing markets are more strict, and failing to meet a commitment can lead to disqualification. Whilst a reservation margin was already applied to account for said strict limits, the simulated response for October 5th shows that forecast deviations can still cause problems where the renewable plant owner cannot meet balancing capacity commitments.

Figure 5.1 presents the generation forecasts, and the minimum bid requirements. Between the hours of 13:00 and 14:00 the forecast 1 suggests generation much higher than the actual value, which is the information available in Stage 1, when available power is committed in the aFRR balancing capacity market. A capacity commitment could be made at that stage, given that the forecasted generation clears the minimum aFRR bid volume even with the reservation margins imposed, but the real generation drops below the minimum bid requirement, which suggests that said bid

could not be honoured, highlighting the risks of participation in balancing markets with PV forecasting tools with large inaccuracies.

The possible solutions include increasing the aFRR reservation margin, which will largely reduce the possible revenues, or improving generation forecasts, which might not be feasible. If better forecasts are available in Stage 3, closer to EAM auction periods, than in Stage 1, when capacity market bids are placed, this can suggest an alternative solution that involves abandoning participation in the balancing markets, and limiting bids to the EAM for which forecast are more accurate. In any case, the results presented in Figure 5.1 highlight the challenges of participation in the balancing markets, for which a simpler two stage model, excluding balancing markets, is used to evaluate the performance of PV plants and their curtailment across a lengthier period of historical data.

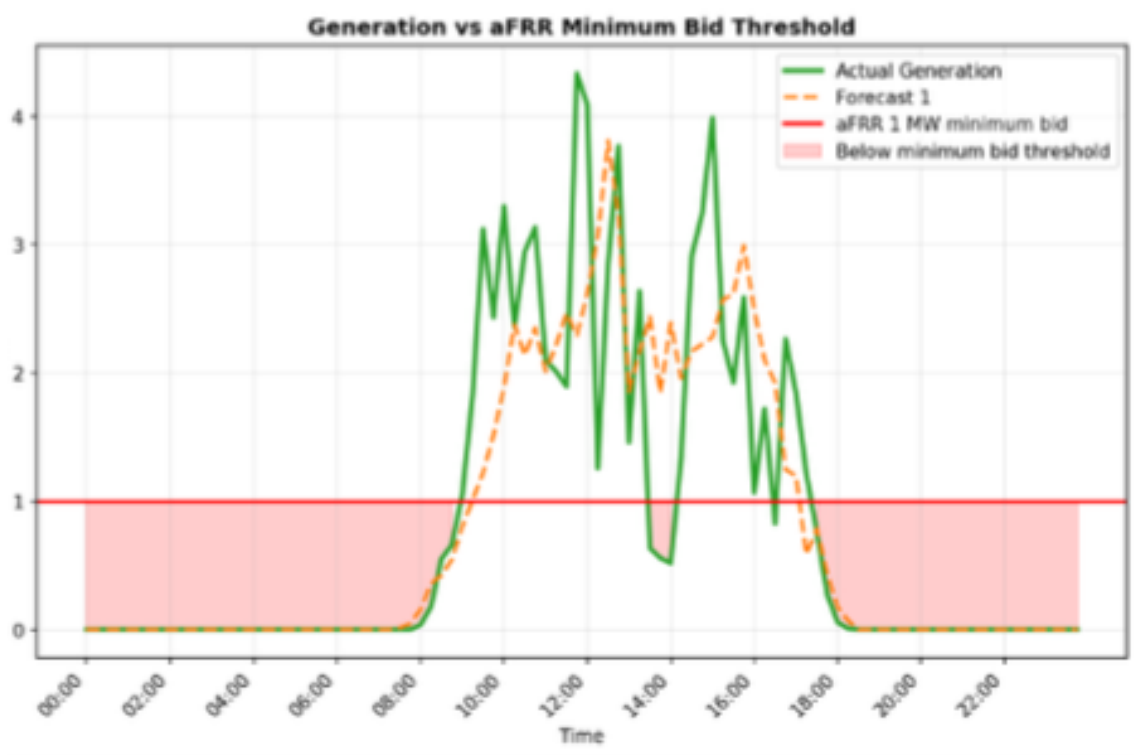


Figure 5.1: aFRR market minimum bid requirement overlaid on top of generation forecast and actual PV output.

5.2 2-Stage Optimization Model Results

This section evaluates the effectiveness of various two-stage models to determine the potential benefits of enhanced governance through a management tool. We focus

exclusively on two-stage scenarios, as most plants currently lack the communication infrastructure, the forecasting accuracy, or fall below the minimum generation threshold required for participation in balancing markets. The scenarios modelled are described in Section 4.5, which describes the market conditions, storage modelling, and optimization strategies that are tested.

The results illustrated in Figure 5.2 indicate a distinct improvement in economic performance as the strategy transitions from a basic DA approach to more sophisticated optimization models that integrate ID trading, curtailment, and storage flexibility. The baseline DA-only scenario generates an annual revenue of approximately €889k, significantly underperforming relative to all optimized configurations. This outcome is expected, as committing all generation in advance precludes the ability to adapt to shifting market conditions, thereby increasing exposure to imbalance costs and forfeiting potential ID market gains. We model a two stage dispatch strategy without aFRR participation, consequently only defensive curtailment is considered, and as such the KPIs discerning offensive and defensive curtailment revenues and volumes are omitted.

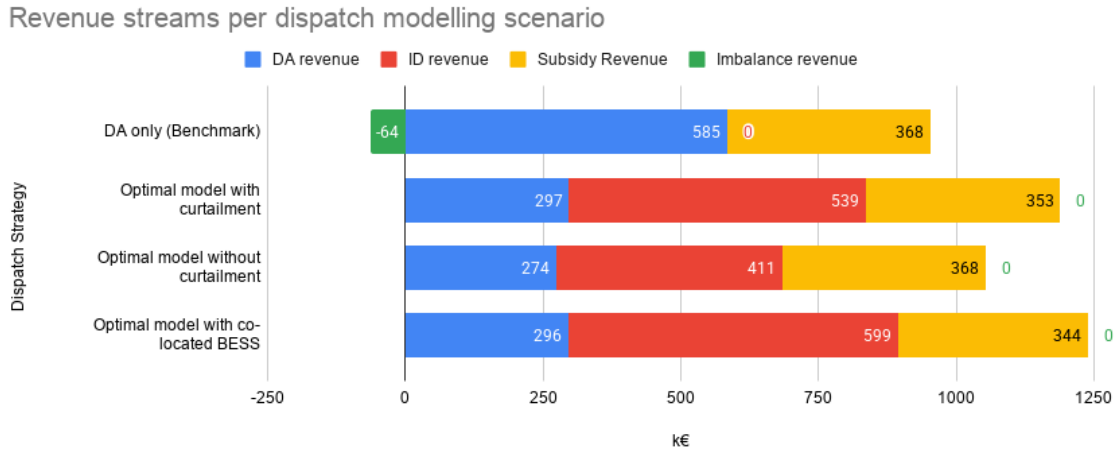


Figure 5.2: Revenue structure and total revenue.

Introducing optimization without curtailment increases revenue to around €1.05M. This improvement is primarily driven by a strong rise in ID revenues, while DA revenues decrease, indicating a strategic shift away from early commitment toward more flexible, real-time market participation. This behaviour is typical for multi-stage optimization frameworks, where decisions are initially based on forecasts and subsequently refined as more accurate information becomes available [50]. It indicates that flexibility and responsiveness are key value drivers in the modelled system.

Allowing curtailment further increases revenues to around €1.19M (+9% compared to the optimized case without curtailment). Curtailment is mainly used as a corrective mechanism to avoid production during low or negative price periods, rather

than as a primary source of value. This is reflected in slightly lower total generation combined with improved revenue quality, particularly in the ID market.

The introduction of BESS leads to the highest revenues across all scenarios. With both storage and curtailment enabled, total revenues increase further to approximately €1.24M. Storage enhances the system’s ability to shift energy temporally, increasing participation in high-value ID moments and further reducing reliance on DA commitments. Notably, a co-located BESS configuration without curtailment is not considered meaningful in the negative price context, as storage alone cannot fully mitigate prolonged negative price exposure without the option to reduce generation. Curtailment and storage therefore act as complementary flexibility options.

Across all scenarios, the EEG subsidy remains a stable component of approximately €340k–€370k per year, providing a consistent revenue floor. This reduces downside risk and dampens the financial impact of negative price periods. Additionally, imbalance costs are largely eliminated in optimized scenarios, indicating that the model treats imbalance avoidance as a strict constraint rather than an economic trade-off. While this simplifies the analysis, it may overestimate the achievable performance in real-world conditions, which would lead to imbalances accross all scenarios, which might reduce revenue, but is expected to affect all strategies equally, as it depends mostly on forecasting accuracy.

Furthermore, the results confirm that each incremental increase in model complexity leads to a significant rise in revenue. These optimization components are complementary; for instance, maximizing the value of a co-located BESS requires a framework that integrates curtailment capabilities with both DA and ID trading. Flexibility both in time (via storage and ID trading) and operational (via curtailment) is consistently rewarded.

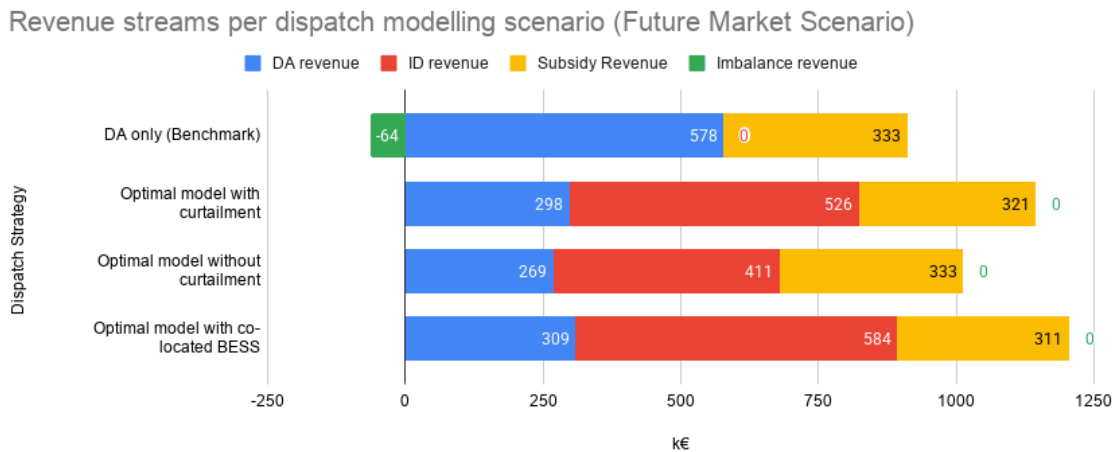


Figure 5.3: Future scenario revenue breakdown.

Current Scenario	Total Revenue [€1000]	DA Revenue [€1000]	ID Revenue [€1000]	Subsidy Revenue [€1000]	Imbalance Revenue [1000€]
DA only (benchmark)	889	585	0	368	-64
Optimal model with curtailment	1.189	297	539	353	0
Optimal model without curtailment	1.053	274	411	368	0
Optimal BESS model (1MWh) with curtailment (avoid imbalance fee)	1.240	296	599	344	0

Table 5.3: Yearly revenue breakdown across scenarios.

Future Scenario	Total Revenue [€1000]	DA Revenue [€1000]	ID Revenue [€1000]	Subsidy Revenue [€1000]	Imbalance Revenue [1000€]
DA only (benchmark)	848	578	0	333	-64
Optimal model with curtailment	1.145	298	526	321	0
Optimal model without curtailment	1014	269	411	333	0
Optimal BESS model (1MWh) with curtailment (avoid imbalance fee)	1.205	309	584	311	0

Table 5.4: Yearly revenue breakdown across scenarios with future negative prices.

To conduct a stress test, the system was evaluated against a future price scenario characterized by heightened volatility and a greater frequency of negative-price hours and grid mandated curtailment. This scenario, detailed in Section 4.5.1, reflects expected long-term market trends and serves to assess the resilience of various governance models. The results are shown in Figure 5.3, and show that overall trends remain largely consistent with present day scenarios, with similar reductions of 3-4% in revenue across all scenarios. This suggests that the rise in negative-price hours does not fundamentally shift dispatch decisions, and that strategies that are optimal nowadays will remain optimal in the future. However, the increased presence of negative price periods and the heightened volatility reduce the profits achievable by the plant owners, and will likely make profit margins more tight, increasing the relevance of all mechanisms to enlarge revenue streams, whether it is economic curtailment, multi market optimization, or storage co-location.

Looking at curtailment volumes, the result from the annual optimization model shown in Table 5.2 indicate a curtailment rate of 13.23% under current market conditions, which is high enough to suggest storage should be implemented under the current system conditions. Since curtailment mainly occurs during midday hours, when renewable output peaks, battery storage appears to be a relevant flexibility option for absorbing excess generation and shifting it to more valuable periods. This

	Curtailment Rate	Curtailed Energy
Optimization with curtailment, current prices	13.42%	1354 MWh
Optimization with curtailment, current prices and battery	12.46%	1257 MWh
Optimization with curtailment, future prices	17.06%	1720 MWh
Optimization with curtailment, future prices and battery	15.81%	1594 MWh

Table 5.5: Curtailment rates and volumes across modelled scenarios.

is highlighted in the curtailment share with a co-located BESS, with a curtailment rate of 12.46%, a near 100 MWh decrease in the entire year. In this study, the battery capacity was not optimized, but fixed at 1 MWh, which explains the relatively low curtailment reduction, which justifies larger BESS installation. However, the large increase in revenue and the reduction in curtailment support the case for battery storage as a tool to reduce curtailment in a way that supports the revenue of PV plant owners.

Furthermore, future scenarios with increased price volatility and negative hours lead to large increases in the expected share of curtailment, growing by 3.64%. Whilst this points towards increasing problems for optimal RE, the results also show that the impact of storage in mitigating curtailment will also grow, with the same BESS reducing the total curtailment share by 1.25% in the future scenarios, compared to a lower 0.96% reduction under current prices.

These results point towards the large impact of economic curtailment, as a revenue raising tool for RE dispatchers, and simultaneously a limitation for decarbonization of Germany's energy system, significantly reducing PV plant power generation. Furthermore, the impact of future market conditions seems to suggest the problem will increase in magnitude, with more curtailment, and tighter profit margins for PV plant owners. However, the relevance of co-located storage as a means to increase revenue and reduce curtailment will become more significant, hinting at electrical storage as a potential solution to increase the output of installed RE power plants.

Chapter 6

Conclusion

In this project, we successfully developed and implemented a multi-stage optimization model to evaluate the strategic value of economic curtailment for solar PV assets in the German electricity market. The final tool simulates an optimal dispatch strategy across several markets, namely the balancing capacity, day-ahead, and intraday markets. The model was tested with real data from the German market, and simulated future scenarios that aim to reflect the state of the energy system in years to come. The results demonstrated that economic curtailment is not just a technical necessity, but can also be a proactive financial strategy for TokWise and its clients, substantially increasing revenue.

The proposed approach combines existing methodologies for a multi-stage and multi-market optimization, which relies on widely available MILP solvers to simulate the dispatch schedule of a RE plant owner. Through the multi-stage approach, forecast uncertainty is modelled to realistically simulate the behaviour at different decision stages across a timeline with improving forecast reliability. The 3-stage model developed was run with real market and PV production data from October 2025. The results suggest that participation in aFRR markets is not possible with the current forecast uncertainty, and whilst several suggestions are made to enable participation in the lucrative balancing markets, ultimately a 2-stage formulation is devised, which excludes the latter markets and optimizes across DA and ID segments of the spot market. This simplified model is benchmarked against several strategies; a naive dispatch strategy bidding all energy only in the DA market, a DA and ID optimizer that does not allow curtailment, and the refined version of the 2-stage model which includes a co-located BESS.

The results suggest that economic curtailment is large, and can reach around 13-17% of total possible generation. This curtailment can improve dispatcher's revenue by close to 10%, an improvement of similar magnitude to the one observed when upgrading a DA-only trading strategy to a multi-market dispatch orchestration correcting traded positions in the ID market. This improvement is also greater than the revenue increase that can be expected from a small BESS co-location, which confirms that economic curtailment is a strong and necessary strategy to maximize revenue for PV plant owners, sometimes even more relevant than the traditional improvements through dispatch optimization. At the same time, the introduction of a relatively small electrical storage of 1MWh reduced curtailment rates by nearly 100 MWh per year, suggesting that such a strategy is effective in raising revenue and preventing the waste of uncontrollable renewable energy generation, particularly in future scenarios with curtailment rates well above 13%.

To simulate the effects of the rapid changes in Germany's energy system, we stress

tested the model under future scenarios with increased levels of price volatility, and more prevalent negative price periods. The results suggests that the revenues for PV plant owners are expected to reduce, and the incentives for curtailment will likely increase due to tighter profit margins. Consequently, economic curtailment is expected to increase by over 3.5%, which hints to a growing problem, that is likely to be tackled by PV producers with electrical storage. Ultimately, these results suggest that curtailment is likely to remain a pressing issue, which if addressed could unlock massive potential, increasing the production of Germany’s future solar energy generation park by a theoretical maximum of 17%, without the need to invest in more plants.

For our industry partner, TokWise, and other organizations in this field, this project demonstrates a micro-level rational for how individual market participants respond to complex price signals. The model offers an initial blueprint for optimizing trading strategies that can scale to broader market-wide policy designs. As renewable energy capacity and market share continue to grow and periods of oversupply become more frequent, the ability to strategically manage generation through economic curtailment will be essential for maintaining the financial viability of renewable portfolios while supporting grid stability. Furthermore, this project findings justify an increased attention to co-located storage, as a solution to increase revenue and maximize the usable output of RE power plants.

In summary, this project concludes that economic curtailment is not only a growing problem of substantial magnitude, but is a vital tool for renewable energy traders today and in the future. To fully utilize the available RE production, significant efforts must be placed to replace the revenue streams currently unlocked by economic curtailment with alternative sources of revenue. This will likely require massive efforts from multiple stakeholders, ranging from the adoption of electrical storage by plant owners, to broader system wide solutions, such as market reforms and investments in energy infrastructure that balance the small scale volatility of RE production with wider portfolios of vast geographical expanses, and will enable a more sustainable and resilient European energy system.

Appendix A

Project Management

This appendix contains additional project management information. Figure A.1 is a Gantt chart showing the project timeline. Table A.1 outlines the project stakeholders and their roles. Table A.2 describes the organization of this project into four stages consisting of nine work packages and provides a description of each.

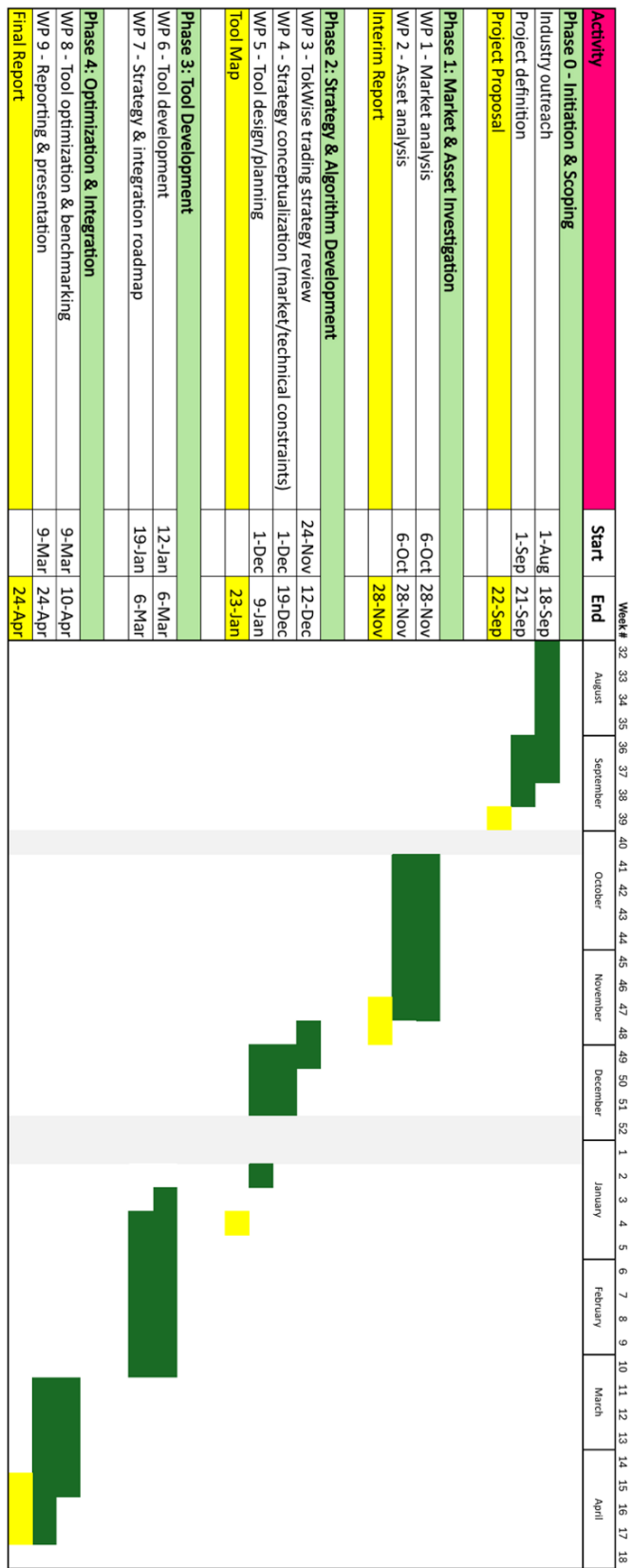


Figure A.1: Gantt Chart providing information on the project’s work packages and its execution timeline.

Group	Name	Organization	Role
Student team	Max Concellon	KTH	WP6 leader
	Shuai Feng	TU/e	WP7 leader
	Mart van Gassel	TU/e	WP4, WP8 leader
	Michelangelo Inzerilli	KTH	WP2 leader
	Matilda Koffa	AGH	WP3 leader
	Emily Larson	IST	Coordinator, WP9 leader
	Carlos Orduña	AGH	WP1 leader
	Stijn Plekkenpol	TU/e	WP5 leader
Industry partner (TokWise)	Julien Marcenac		
	Nadia Daneva		
	Ivaylo Andreev		
Academic supervisor	Duarte de Mesquita e Sousa		
Alumni advisor	Nina Jabłońska		

Table A.1: Project stakeholders.

WP	WP Name	Start	Finish	Description
0	Project coordination	June	April	Project development, stakeholder engagement, report writing, team management
Phase 1: Market & Asset Investigation				
1	Market analysis	6 Oct	14 Nov	Literature review of economic curtailment, how it is currently being implemented, future plans, review of current curtailment strategies and results, interviews with experts
2	Asset technical analysis	6 Oct	14 Nov	Literature review of the technical considerations and limitations of curtailment for wind and solar PV assets
Phase 2: Strategy & Algorithm Development				
3	TokWise trading strategy review	10 Nov	12 Dec	Review of current TokWise trading strategies, clients, products; identification of gap or opportunity for our model to fill
4	Strategy conceptualization	1 Dec	19 Dec	Market and technical constraints for the tool, boundaries for its operation, how it can fit into the TokWise ecosystem and provide value to clients
5	Tool design/planning	1 Dec	9 Jan	Requirements, inputs, outputs, structure of tool, KPIs, plan for how to actually develop it
Phase 3: Tool Development				
6	Tool development	12 Jan	6 Mar	Basic working model of the tool
7	Strategy and integration roadmap	12 Jan	6 Mar	Analysis of how this tool can be used in TokWise's portfolio, market impact
Phase 4: Optimization & Integration				
8	Tool optimization and benchmarking	9 Mar	10 Apr	Optimized tool and analysis of its performance against benchmarks and KPIs
9	Reporting and presentation	30 Mar	24 Apr	Final report and presentation

Table A.2: Work packages.

Appendix B

Obtaining Single Value Estimations for the Optimal aFRR Bidding Price

The calculation of an optimal aFRR energy activation market bid price is done according to the five-step procedure outlined in this appendix. It is inspired by the activation-simulation approach of Loeach, Rominger et al. [36] and the activation-duration formulation used in Gärtner et al. [37].

The five steps are:

1. Build the merit order list from anonymous bids.
2. Determine activation demand each minute.
3. Compute the probability of bid activation for each bid price level.
4. Compute the optimal bidding price.
5. Estimate the expected activation compensation in EUR of revenue per MWh bid.

Step 1 – Build the merit order list

The Merit Order Lists (MOLs) are generated based on the anonymous energy activation market bids by cumulating the reserve power of the single bids sorted by the bids' offered energy prices. This forms a supply curve with the x-axis being cumulative reserve power (MW), and the y-axis as energy price (EUR/MWh). For clarity, the sign of the bids is inverted, such that a positive value corresponds with the compensation received from curtailing a given energy, also with a positive sign. An example of such a MOL is shown in Figure B.1 below.

Step 2 – Determine activation demand each minute

Calculate minutely reserve demand (how much balancing energy the TSO needs per each period), separate for positive and negative reserve. Then, the activated energy S_t (MW) represents how much balancing energy the TSO requires in each activation period. Aggregation from 4-second activation periods into minute-long periods follows the approach found in the literature [36]. This approximation reduces computing power and better matches the final TSO's activation data, which, due

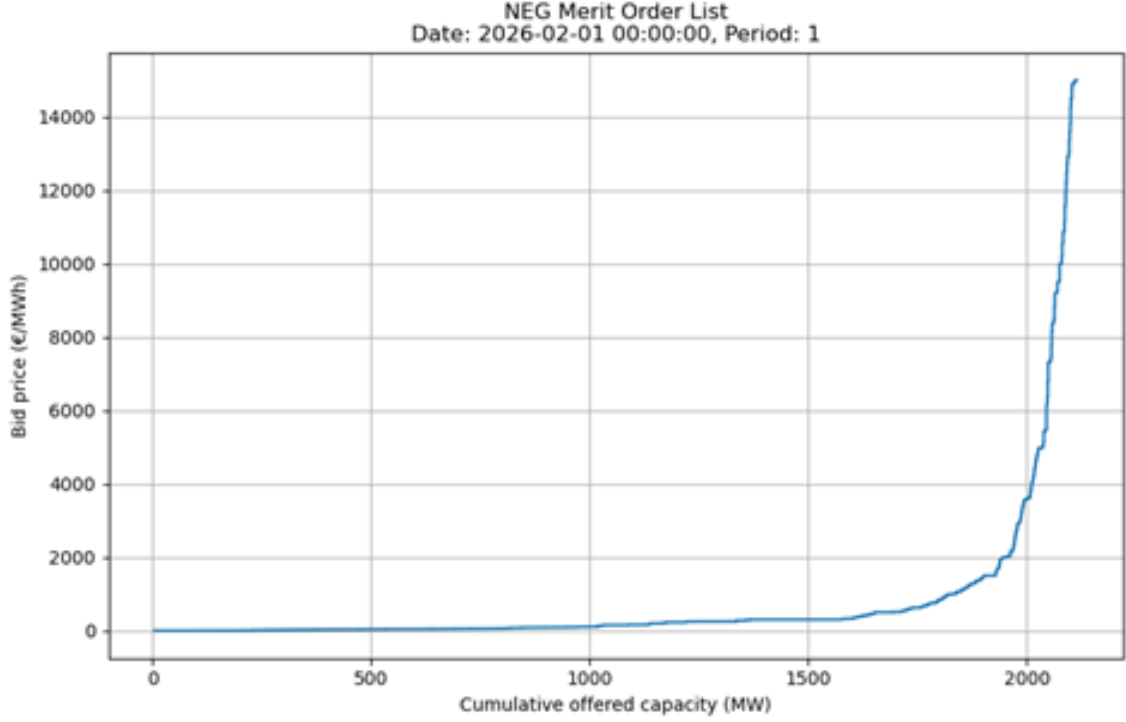


Figure B.1: Example of a Merit Order List generated from anonymous bid information for a 15 -minute interval.

to the ramp up and ramp down needs of reserves, generally leads to activations in the order of minutes rather than seconds granularity, making this an acceptable simplification of the process.

The energy price MOLs and the calculated minutely SCR demands are matched, to allow determining the highest paid energy prices for each minute either for positive or negative reserve. Once activation demand is met, the last bid activated determines the highest paid price. This highest paid price represents the marginal activation price for that minute long period. Hence,

$$P_{max} = \min\{p : \Psi(p) \geq S_t\} \quad (\text{B.1})$$

where S_t represents the activated volume in MW at each instant t , and $\Psi(p)$ represents the cumulative MW in the merit order list up to price level p , and P_{max} is the highest price needed to supply each activation.

Step 3 – Compute the activation probability (activation share)

A bid is activated if its energy price (p) is less than or equal to the highest activated price within each time interval, $p \leq P_{max,t}$. Over the entire 15-minute interval for which the bid is applicable, calculate the probability of activation as $\bar{P}(activated|p|)$, which is the number of time steps activated divided by the total time steps. For an entire period with several activation intervals N ;

$$\bar{P}(activated|p|) = 1/N \sum_{t=1}^N 1 * \{p \leq p_{max,t}\} \quad (B.2)$$

where $\bar{P}(activated|p|)$ is the estimated probability of activation of a given bid at price p , where N are the total number of bid activation intervals in the reference bidding period, and $\{p \leq p_{max,t}\}$ is an indicator function, which is 1 if a bid is activated, meaning the bid price p is below the maximum bid price p_{max} of that interval, and is zero otherwise. An example of the different possible values for a given interval is shown in figure B.2.

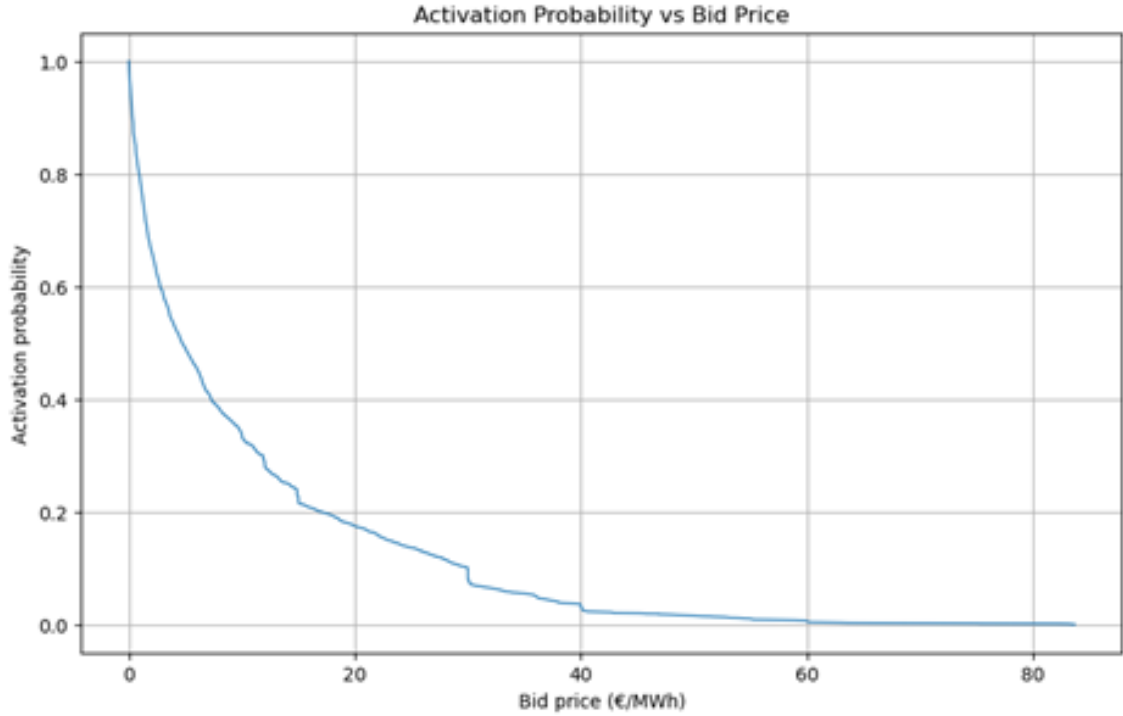


Figure B.2: Example of the activation probability for aFRR EAM bids in an example 15-minute interval.

Step 4 – Compute the optimal bidding price

To find the optimal p (bid price), possible prices are restricted to discrete candidates matching the unique historical bids found in the data. For each period t , compute the curve and then chose the optimal bid price p as:

$$p_{optimal} = \arg_p \max(0.25 * p * \bar{P}(activated|p)) \quad (B.3)$$

That is, choose the p that maximizes the expected pay-as-cleared activation compensation per MW per 15-min interval, based on the probability of activation $\bar{P}(activated|p)$.

Step 5 – Estimate expected activation compensation

Using the activation probability estimated in Step 4, the expected activation revenue from the Energy Activation Market (EAM) can be computed for each 15-minute interval. Given m (MW), the offered capacity that should be a decision variable in the real optimization model, we calculate the compensation for an m of 1 MW, which can then be extrapolated to other quantities;

$$ExpectedCompensation \frac{EUR}{MWh} = m_k * \Delta t * p_{optimal} * \bar{P}(activated|p_{optimal}) \quad (B.4)$$

Where m_k is the reserve capacity in MW offered in interval k , Δt is the duration of the interval (generally 0.25 h), $p_{optimal}$ is the optimal bid price in euros per MWh, and $\bar{P}(activated|p_{optimal})$ is the activation probability for the chosen optimal bid price.

The resulting output Excel dataset contains activation probability functions and expected pay-as-cleared compensation derived from historical EAM marginal prices. These provide quantitative estimates of expected EAM bid compensation that can be directly incorporated into the objective functions of optimization models at all stages. The RE trader is expected to have information on these values, and their derived forecasts. From this, it is rational to bid at the identified optimal bidding price for each period, such that the bidding price becomes a fixed parameter, and the optimization function simply decides on the chosen volume bid at the expected aFRR EAM price level.

Appendix C

Input & Decision Variables for Optimization Model

In this appendix, the input, decision, and auxiliary variables for each of the three model stages are described in six tables.

Stage 1: Balancing Capacity Market

- Table C.1 contains the input variables for Stage 1.
- Table C contains the decision and auxiliary variables for Stage 1.

Stage 2: Day Ahead Market

- Table C.3 contains the input variables for Stage 2.
- Table C contains the decision and auxiliary variables for Stage 2.

Stage 3: Intraday and Balancing EAM

- Table C.5 contains the input variables for Stage 3.
- Table C contains the decision and auxiliary variables for Stage 3.

Input Variable	Description	Type
DA_p_f2	Price for the given time period in the DA market, as forecasted under forecast 1 (EUR/MWh).	vector
ID_p_f2	Price for the given time period in the ID market, as forecasted under forecast 1 (EUR/MWh).	vector
EEG_p_actual	EEG subsidy price applied to a given period, including subsidy neutralization rule for negative price periods (EUR/MWh).	vector
EAM_p_f2	Calculated compensation expected for aFRR negative EAM, under forecast 1 (EUR/MWh).	vector
BCap_Neg_p_f2	Forecasted compensation for aFRR negative capacity market (EUR/MW).	vector
capacity	RE power generation installed capacity (MW)	single value
aFRR_reservation_margin	Maximum expected forecast deviation for RE generation, fixed at 0.67 (67%).	single value
Generation_f2	Expected RE generation (in MW) under forecast 2.	vector

Table C.1: Input variables for Stage 1.

Decision Variable	Description	Bounds
DA_stage1	DA scheduled quantity for delivery period t . Represents the expected amount of electricity (in MW) sold in the DA market under the follow-the-money convention.	continuous $capacity \geq DA_t \geq 0$
ID_stage1	ID scheduled quantity for delivery period t . Represents the expected amount of electricity (in MW) sold in the ID market.	continuous $capacity \geq ID_t \geq 0$
aFRR_N_stage1	Offered aFRR volume committed in the capacity market for delivery period t . Represents the amount of electricity (in MW) that is committed to be bid in the EAM auction.	continuous $aFRR_N_stage1_t \geq 1$ or 0 and $aFRR_N_stage1_t \leq (DA_t + ID_t) * aFRR_reservation$
aFRR_N_stage1_binary	Auxiliary variable used to ensure aFRR bids have a minimum volume of 1MW.	binary
aFRR_block	Auxiliary variable to enforce continuity constraint, and ensure bids in aFRR capacity market are placed in 4 h blocks.	Same bounds as aFRR_N_stage1 with 6 different possible values per day

Table C.2: Decision and auxiliary variables for Stage 1.

Input Variable	Description	Type
DA_p_f1	Price for the given time period in the DA market, as forecasted under forecast 1 (EUR/MWh).	vector
ID_p_f1	Price for the given time period in the ID market, as forecasted under forecast 1 (EUR/MWh).	vector
EEG_p_actual	EEG subsidy price applied to a given period, including subsidy neutralization rule for negative price periods (EUR/MWh).	vector
EAM_p_f1	Calculated compensation expected for aFRR negative EAM, under forecast 1 (EUR/MWh).	vector
BCap_Neg_p_actual	Actual compensation for aFRR negative capacity market (EUR/MW).	vector
capacity	RE power generation installed capacity (MW)	single value
aFRR_reservation_margin	Maximum expected forecast deviation for RE generation, fixed at 0.67 (67%).	single value
Generation_f1	Expected RE generation (in MW) under forecast 1.	vector
aFRR_N_fixed	The aFRR capacity market volume committed in Stage 1 (MW)	vector

Table C.3: Input variables for Stage 2.

Decision Variable	Description	Bounds
DA_stage2	DA scheduled quantity for delivery period t . Represents the expected amount of electricity (in MW) sold in the DA market.	continuous $capacity \geq DA_t \geq 0$
ID_stage2	ID scheduled quantity for delivery period t . Represents the expected amount of electricity (in MW) sold in the ID market.	continuous $capacity \geq ID_t \geq 0$
aFRR_N_stage2	Expected aFRR volume to be bid in EAM market for delivery period t . Represents the amount of electricity (in MW).	continuous $aFRR_N_stage2_t \geq 1$ or 0 and $aFRR_N_stage2_t \leq (DA_t + ID_t) \times aFRR_reservation_margin$
aFRR_N_stage2_binary	Auxiliary variable used to ensure aFRR bids have a minimum volume of 1MW.	binary

Table C.4: Decision and auxiliary variables for Stage 2.

Input Variable	Description	Type
DA_p_actual	Price for the given time period in the DA market (EUR/MWh).	vector
ID_p_actual	Price for the given time period in the ID market (EUR/MWh).	vector
EEG_p_actual	EEG subsidy price applied to a given period, including subsidy neutralization rule for negative price periods (EUR/MWh).	vector
DA_fixed_stage3	Traded energy in the DA market, as decided in Stage 2 (MW)	vector
BIGM	Auxiliary variable to ensure imbalance exclusivity.	fixed value = 100
c_pos	Positive imbalance penalty (EUR/MWh)	fixed value = 100 000
c_neg	Negative imbalance penalty (EUR/MWh)	fixed value = 100 000
EAM_p_actual	Actual compensation for aFRR negative EAM (EUR/MWh)	vector
BCap_Neg_p_actual	Actual compensation for aFRR negative capacity market (EUR/MW)	vector

Table C.5: Input variables for Stage 3.

Decision Variable	Description	Bounds
ID_stage3	ID scheduled quantity for delivery period t . Represents the expected amount of electricity (in MW) sold in the ID market.	continuous $-capacity \leq ID_t \leq capacity$
imb_pos_stage3	Positive imbalance willingly incurred representing the positive difference between delivered energy and traded energy in ID and DA markets (in MW).	continuous 0 lower bound
imb_neg_stage3	Negative imbalance willingly incurred representing the negative difference between delivered energy and traded energy in ID and DA markets (in MW).	continuous 0 lower bound
delta_imb_stage3	Auxiliary variable to ensure imbalance direction exclusivity, to avoid incurring positive and negative imbalances.	binary
aFRR_N_stage3	Actual aFRR volume to be bid in EAM market for delivery period t . Represents the amount of electricity (in MW).	continuous $aFRR_N_stage3_t \geq 1$ or 0 and $aFRR_N_stage3_t \leq (DA_t + ID_t) \times aFRR_reservation_margin$
aFRR_N_stage3_binary	Auxiliary variable used to ensure aFRR bids have a minimum volume of 1MW.	binary

Table C.6: Decision and auxiliary variables for Stage 3.

Appendix D

3-Stage Optimization Results

D.1 First KPI results analysis

The KPI results reveal that offensive curtailment through the aFRR market (KPI 3 = €1,245.81) contributed 98.0% of total revenue, comprising the aFRR balancing capacity (BCap) capacity payment of €1,200.94 and the EAM activation revenue of €44.86. The remaining 2.0% (€25.71) came from energy market trading. This finding demonstrates the transformative impact of aFRR market access, when the asset is large enough to participate, the balancing capacity payment completely dominates the trading economics. Without aFRR, the Optimized Model earns €25.71 against the Benchmark Model's €-13.36 on energy markets alone, an improvement of €39.06. With aFRR included, the total advantage over the benchmark is €1,284.88.

All of the EEG Market Premium reported in Table D.1 (€8.74) are generated in positive price hours, since under §51 of the EEG, the market premium is set to zero for any settlement period in which the DA price is at or below zero, a provision designed to eliminate the incentive to generate during system oversupply. On 5 October 2025, this restriction applied to 70 of the 96 quarter-hourly periods. The €8.74 reported in Table D.1 represents EEG premium revenue earned exclusively during the remaining 26 non-negative price periods, during which the §51 restriction did not apply and the full market premium was receivable.

D.2 Cumulative Revenue: Optimized vs. Benchmark

Figure D.1 presents the cumulative energy market revenue across all 96 quarter-hourly periods of the analysis day for both the Optimized Model and the Benchmark Model, in which energy is bid in the DA market according to the forecast available at that time, and no further trading is carried out. It should be noted that the aFRR BCap revenue of €1,200.94 is not reflected in this figure as it is settled separately at Stage 1 and does not flow through the energy market settlement. The true total improvement including aFRR is €1,284.88, making the energy market comparison shown here only a partial picture of the full advantage of the optimized strategy.

Both models record zero revenue through the overnight and early morning hours when there is no solar generation. From approximately 07:30, as daylight generation becomes available, the two models diverge. The Optimized Model begins

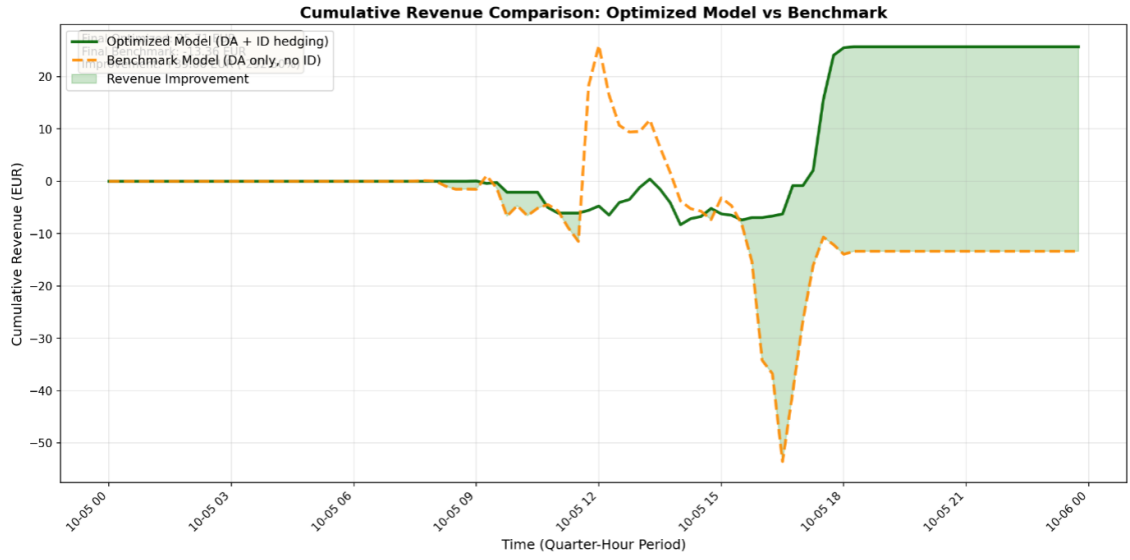


Figure D.1: Cumulative revenue optimized vs benchmark.

Model	Revenue
Optimized Model	€25.71
Benchmark Model	€-13.36
Improvement	€39.06 +292.50%

Table D.1: Final cumulative revenue.

Market	Revenue	Volume	VWAP
Day-Ahead Market	€-7.06	2.07 MWh	€-3.41/MWh
Intraday Market	€32.67	14.44 MWh	€2.26/MWh
EEG Market Premium	€8.74	—	€0.34/active period
Imbalance Settlement	€-8.64	—	—
Total	€25.71		

Table D.2: Energy market revenue and volume statistics.

accumulating revenue steadily through the ID market during the solar generation window, ending the day at €25.71.

The Benchmark Model, whose strategy is to sell all forecasted generation in the DA market and accept all resulting imbalances without ID correction, immediately begins losing value at the start of the day. Committing 18.78 MWh to the DA market on a day where 70 out of 96 periods had negative DA prices, and where 38 out of 42 generation periods specifically had negative DA prices, generates a DA loss of €-45.89. The benchmark recovers partially through the late-evening price spike when DA prices reached €49.41/MWh and through an imbalance settlement of €23.79, positive on this day only because actual generation happened to exceed the DA commitment on balance, but ends the day at €-13.36. The €39.06 gap between the two models on energy markets confirms that sequential optimization, by avoiding negative DA price exposure and routing generation through the higher-priced ID market, substantially outperforms passive DA commitment.

It is worth noting that the Benchmark Model temporarily exceeds the Optimized Model’s cumulative revenue around 12:00, reaching approximately €24 before falling sharply. This occurs because the benchmark commits all forecasted generation to the DA market regardless of price, in the few midday periods where DA prices were briefly positive the benchmark captured a larger DA revenue than the optimized model which had already routed generation to ID. This temporary advantage is entirely coincidental and reflects the benchmark accidentally benefiting from a small number of positive DA periods rather than any strategic superiority. As DA prices turned sharply negative through the afternoon and evening the benchmark suffered severe losses, confirming that the optimized model is designed to maximise end-of-day total revenue across all markets rather than outperform the benchmark in every individual period.

D.3 Revenue Breakdown by Market

Figure D.2 shows the composition of energy market revenue. The ID market was the dominant positive contributor at €32.67, representing the core of the defensive strat-

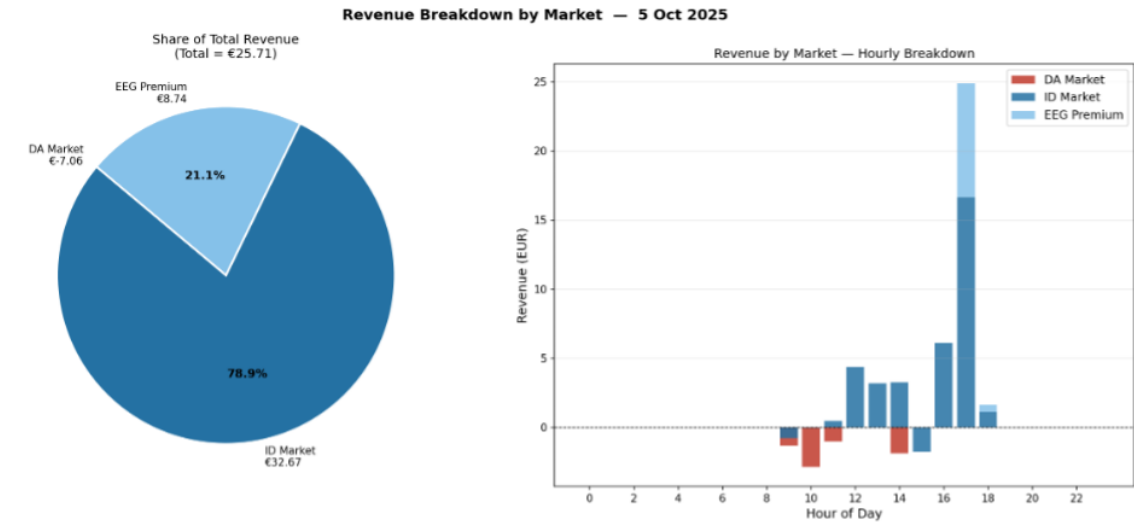


Figure D.2: Cumulative revenue optimized vs benchmark.

egy: routing generation through the ID market where the volume-weighted average price (VWAP) realized was €2.26/MWh, rather than committing heavily to the DA market which resulted in a realized VWAP of €-3.41/MWh. The EEG premium provided a consistent supplementary revenue stream, averaging €0.34 per period during the 26 non-negative daylight hours it was active. Unlike a simple arithmetic average of market prices, VWAP weighs each price by the specific volume traded during that interval. This metric is critical in high-volatility environments, such as the negative price periods observed, as it reflects the actual realized value for every MWh sold. This model committed 2.07 MWh to the DA market in some afternoon and evening periods where positive DA prices made commitment worthwhile. However, the net DA revenue was negative at €-7.06, reflecting a volume-weighted average DA price of €-3.41/MWh across the committed periods, the timing of some commitments coincided with periods where actual prices turned negative relative to expectations.

The imbalance settlement of €-8.64 reflects 6 negative imbalance periods arising from the interaction between the aFRR capacity obligation locked in at Stage 1 and the ID hedging requirements at Stage 3. As discussed in Sections 5.4.1 and 5.4.3, imbalances are left as a decision variable in Stage 3 and penalised at €100,000/MWh to effectively prohibit deliberate imbalances. These 6 periods represent a structural consequence of aFRR participation, when the aFRR obligation constrains available generation, a shortfall can occur that the ID market cannot fully correct. This risk can only be addressed through more accurate forecasting, more conservative reservation margins, or co-located BESS, as discussed in Section 5.2.1. While emphasized here for aFRR participation, this structural risk is inherent to any DA commitment: forecast errors can lead to shortfalls that cannot be economically recovered if ID prices are unfavorable.

Metric	Stage 2 (Forecast 1)	Stage 3 (Actual)
DA total	2.07 MW	2.07 MW
ID total	12.04 MWh	14.44 MWh

Table D.3: Summary statistics across Stage 2 and Stage 3.

D.4 Optimization Results Across Three Stages

Figure D.3 shows the optimization results across three stages through four subplots. The generation subplot confirms the solar profile with forecast peaking at 3.81 MW and actual generation reaching 4.33 MW, with the asset slightly underforecasted throughout the daylight hours. The DA subplot shows 2.07 MWh committed at Stage 2 and unchanged at Stage 3, confirming the DA commitment was correctly carried forward as a fixed input. The ID subplot shows an increase from 12.04 MWh planned at Stage 2 to 14.44 MWh final at Stage 3, as the additional actual generation above the forecast was absorbed by the ID market at actual prices. The fourth subplot presents the full Stage 3 market allocation — showing generation possible, generation after curtailment, DA commitment, ID trading, and any positive or negative imbalance positions together in a single view, giving a complete picture of how the available generation was distributed across all market outlets at the final settlement stage.

D.5 Decision Comparison: Stage 1 vs. Stage 2 vs. Stage 3

Figure D.4 compares decisions across all three stages through six subplots: generation forecasts, DA prices, DA volumes, ID volumes, aFRR volumes, and ID prices. The aFRR capacity bid of 1.24 MW was committed during the daylight four-hour blocks at Stage 1 and this obligation carries unchanged through Stage 2 and Stage 3, resulting in 8.09 MWh of energy dispatched through the EAM over the day. The DA volume subplot shows that Stage 1 planned approximately 0.5 MWh of DA commitment, which Stage 2 revised upward to 2.07 MWh as Forecast 1 updated the expected market conditions — Stage 3 then carried this commitment forward unchanged, confirming DA was correctly fixed after Stage 2. The ID volume subplot shows a clear progression across all three stages: Stage 1 planned 11 MWh, Stage 2 refined this to 12 MWh, and Stage 3 final reached 14.44 MWh as actual generation exceeded the forecast and the additional volume was absorbed by the Intraday market. The price subplots confirm that the market conditions visible across all three stages — predominantly negative DA prices, positive ID prices during daylight hours, and attractive aFRR BCap prices in the midday blocks — consistently

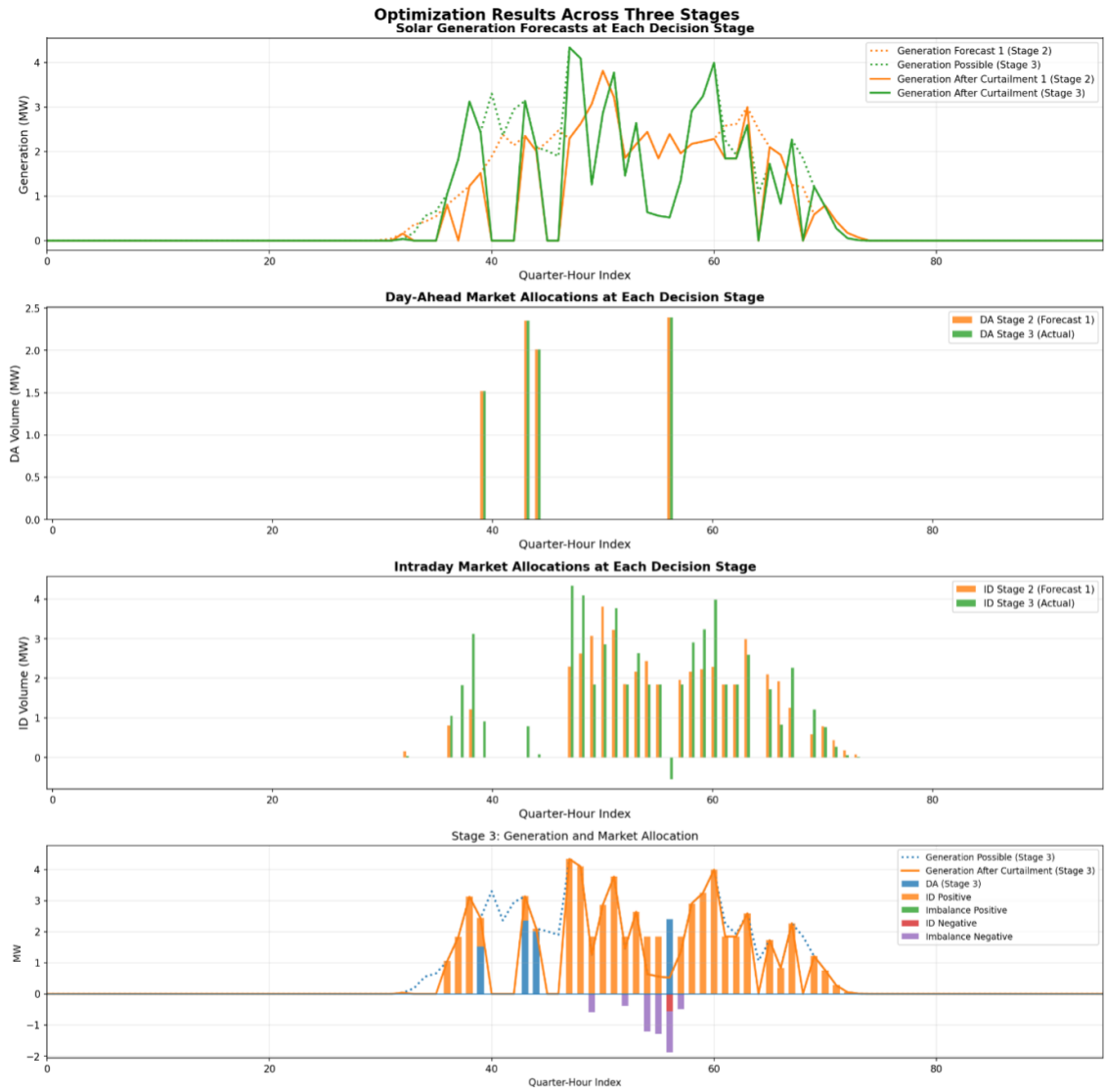


Figure D.3: Optimization results across three stages.

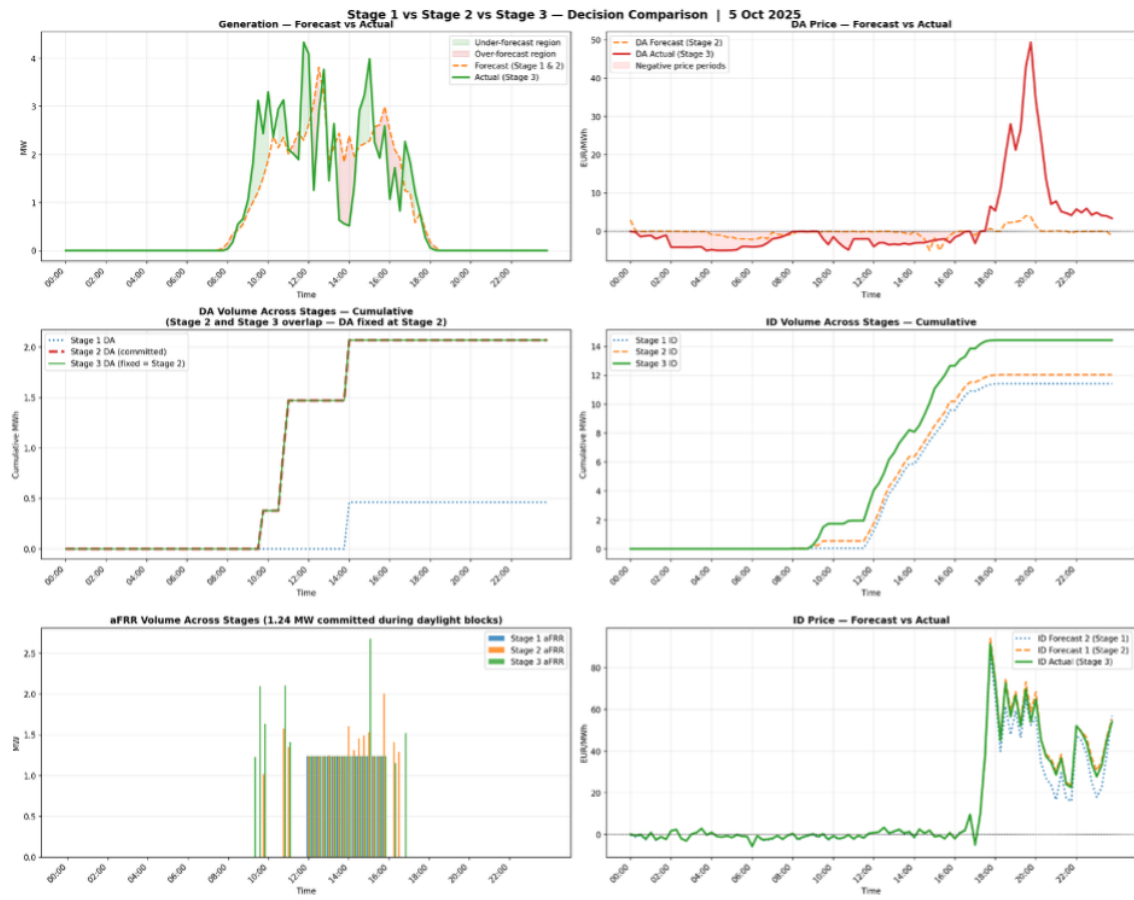


Figure D.4: Decision comparison - Stage 1 vs. Stage 2 vs. Stage 3.

justified the same core strategy at every decision point.

D.6 DA and ID Market Analysis

Figure D.5 presents three subplots examining DA and ID market activity in detail. The revenue and traded volume subplot shows ID revenue concentrated in the solar generation window between approximately 07:30 and 18:00, with DA contributions small and largely negative. Rather than relying on simple daily price averages, the economic superiority of the ID preference is quantified by the actual realised volume-weighted average prices shown in Table D.2, the model achieved an effective ID rate of €2.26/MWh while the DA commitment resulted in a rate of €-3.41/MWh. The cumulative traded volume subplot confirms that the ID market absorbed the majority of the day’s generation, reaching 14.44 MWh by end of day against the DA’s 2.07 MWh. The cumulative price sold subplot shows how each market stream contributed to the total revenue over time — the DA line correctly falls to €-7.06, the ID line rises to €32.67, the EEG premium builds steadily to €8.74, and the imbalance settlement pulls the total slightly negative before the final settlement of €25.71 at end of day.

D.7 Market Prices & Trading Volumes

Figure D.6 integrates generation, DA, and ID activity into three aligned subplots to show how the physical generation profile drove the trading decisions. The top panel shows the generation forecast versus actual solar profile, both rising from zero at sunrise to a peak around midday and falling back to zero by early evening, with actual generation (4.33 MW peak) slightly above the forecast (3.81 MW peak) throughout the daylight hours. The middle panel shows the DA market: price plotted with traded volume bars showing the 2.07 MWh committed across selected periods. The bottom panel shows the ID market: price plotted with traded volume bars corresponding to the 31 periods where the model sold generation into the ID market at prices averaging €12.93/MWh, generating €32.67 in ID revenue. The total actual generation of 19.52 MWh is fully accounted for across four components: 2.07 MWh delivered through DA, 14.44 MWh sold through ID, 4.33 MWh defensively curtailed during negative price periods, and a negative imbalance shortfall of 1.32 MWh across the 6 periods where the aFRR obligation prevented full ID correction , confirming that $2.07 + 14.44 + 4.33 - 1.32 = 19.52$ MWh.

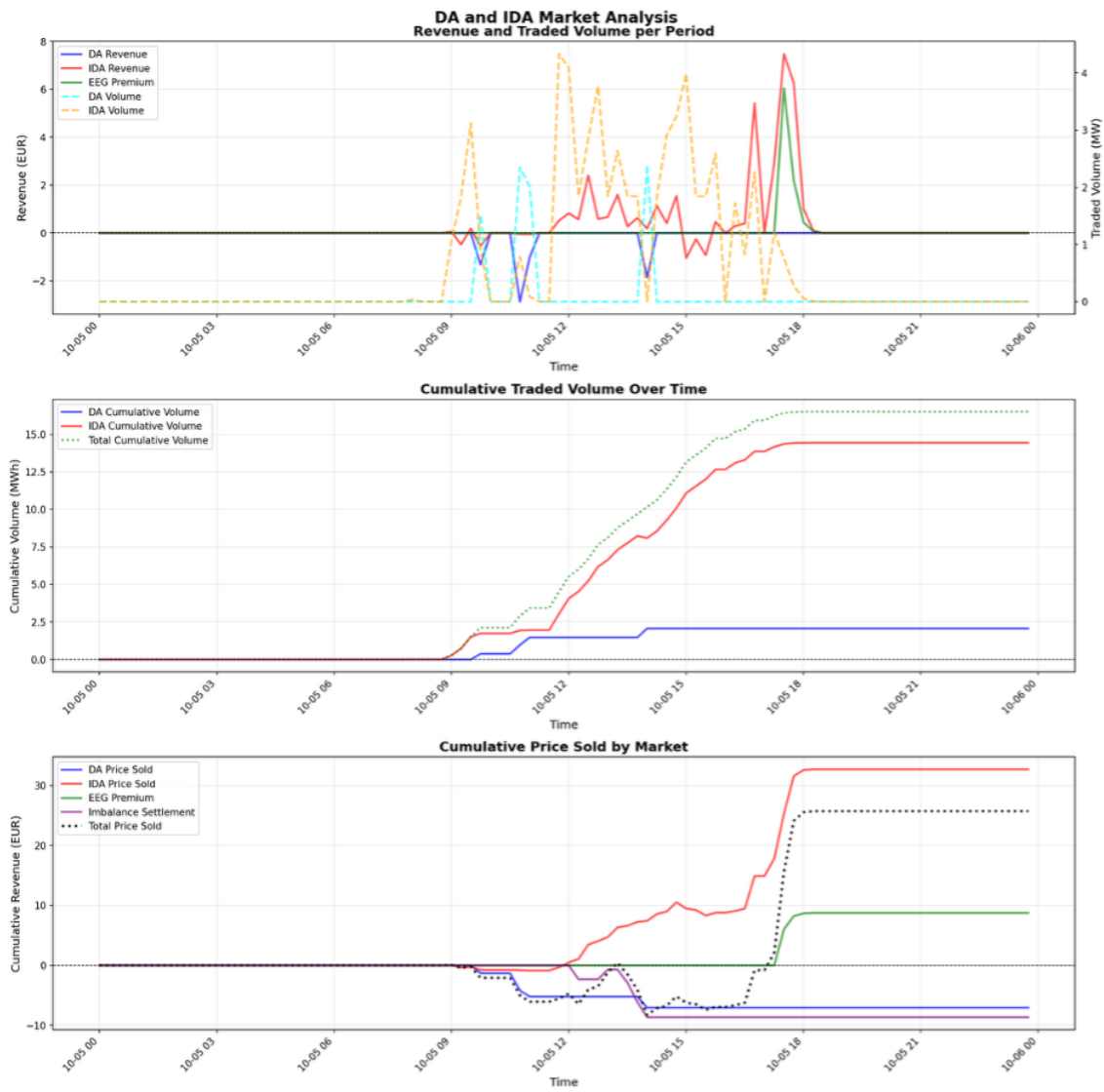


Figure D.5: Day ahead and Intraday market analysis.

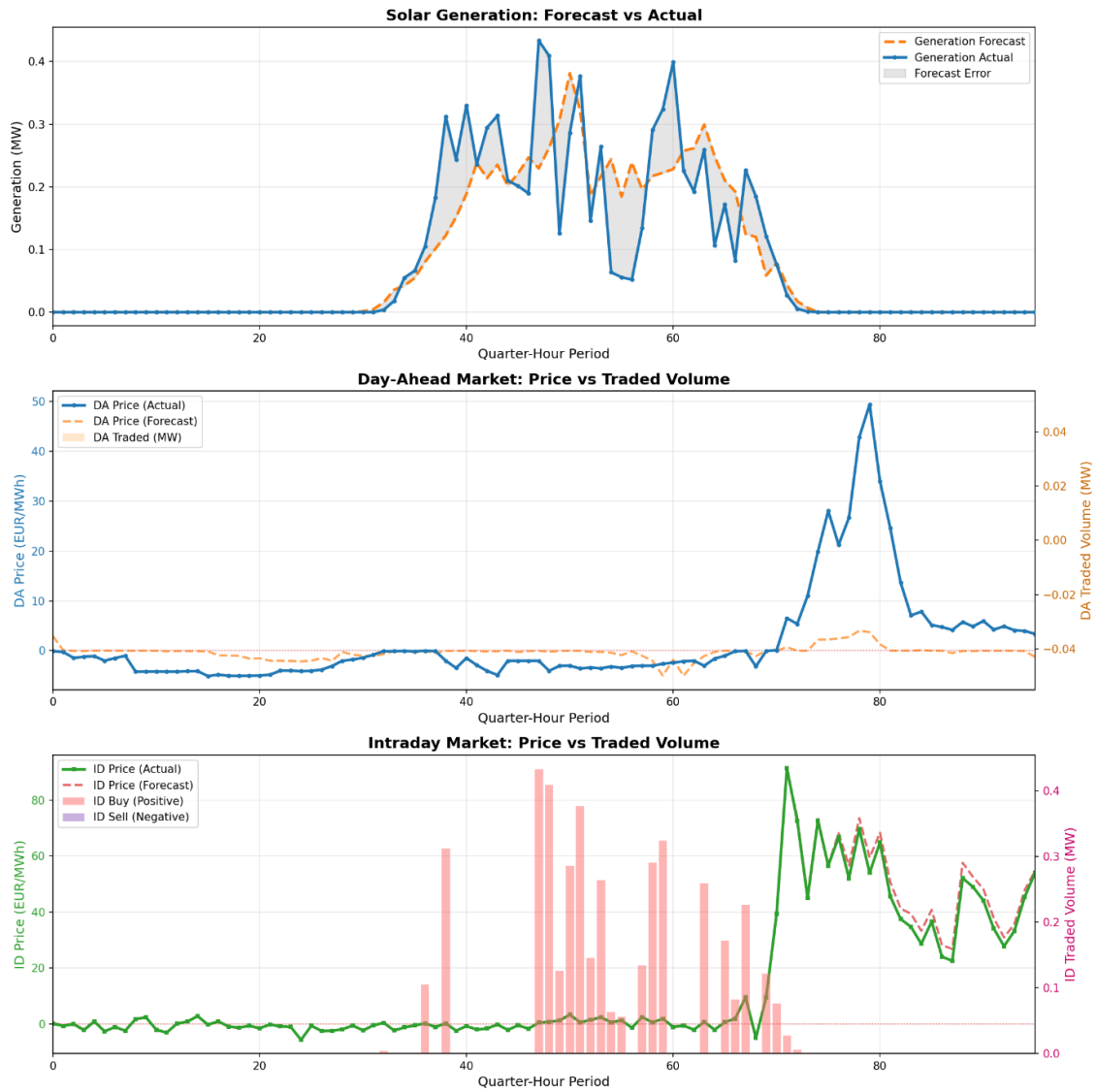


Figure D.6: Market prices vs. trading volumes.

	Values
Generation forecast total	18.78 MWh
Actual generation total	19.52 MWh
Forecast error	0.73 MWh (3.9%)
Total DA volume	2.07 MWh
Total DA revenue	€-7.06
ID buy periods	1 period, 0.14 MWh total
Generation curtailed	4.33 MWh
ID sell periods	31 periods 14.44 MWh total
Total ID revenue	€32.67
Generation Accounting	
DA delivered	2.07 MWh
ID delivered	14.44 MWh
Curtailed	4.33 MWh
Negative imbalance shortfall	-1.32 MWh
Total accounted for	19.52 MWh

Table D.4: Price vs. volume statistics.

D.8 Curtailment Analysis

Figure D.7 presents the curtailment analysis through three subplots: the DA and ID price time series with negative DA periods highlighted, the curtailment volume per period classified by price type, and the financial loss avoided per period through defensive action.

Subplot 1 shows both the DA price (solid blue) and the ID price (dashed orange) across all 96 periods, with negative DA price periods shaded in red. This is important context for interpreting the curtailment decisions: by Stage 3 the DA commitment was already fixed, meaning the optimizer’s curtailment decision was purely a choice between selling remaining available generation on the ID market or withholding it entirely. The subplot shows that during the 12 curtailed periods — all concentrated in the morning and early afternoon solar generation window — both DA and ID prices were low or negative simultaneously, confirming that curtailment was the rational outcome when neither market offered an attractive price.

Subplot 2 shows all 4.33 MWh of physical curtailment classified as negative price period curtailment (red bars), with zero curtailment during positive price periods (no orange bars visible). This confirms the model captured every available revenue opportunity when market conditions were favourable and withheld generation only

when both markets were unattractive.

Subplot 3 quantifies the financial benefit of each curtailment decision, showing the loss avoided per period. The largest single-period saving of approximately €3.00 occurred around 10:30, the period with the largest curtailed volume (0.80 MWh) coinciding with the most negative price of the curtailed window. Total defensive curtailment savings of €10.07 (KPI 5) represent the cumulative losses avoided across all 12 curtailed periods.

Offensive curtailment operates through a fundamentally different pathway and does not appear in this figure. The 8.09 MWh of offensive curtailment volume (KPI 4) represents generation committed for aFRR balancing activation through the EAM, generating €1,245.81 in offensive curtailment revenue (KPI 3). Both defensive and offensive curtailment mechanisms were active on this day, demonstrating the full scope of economic curtailment the model is designed to capture, defensive curtailment protecting against spot market losses on the downside, and offensive curtailment generating balancing market revenue on the upside.

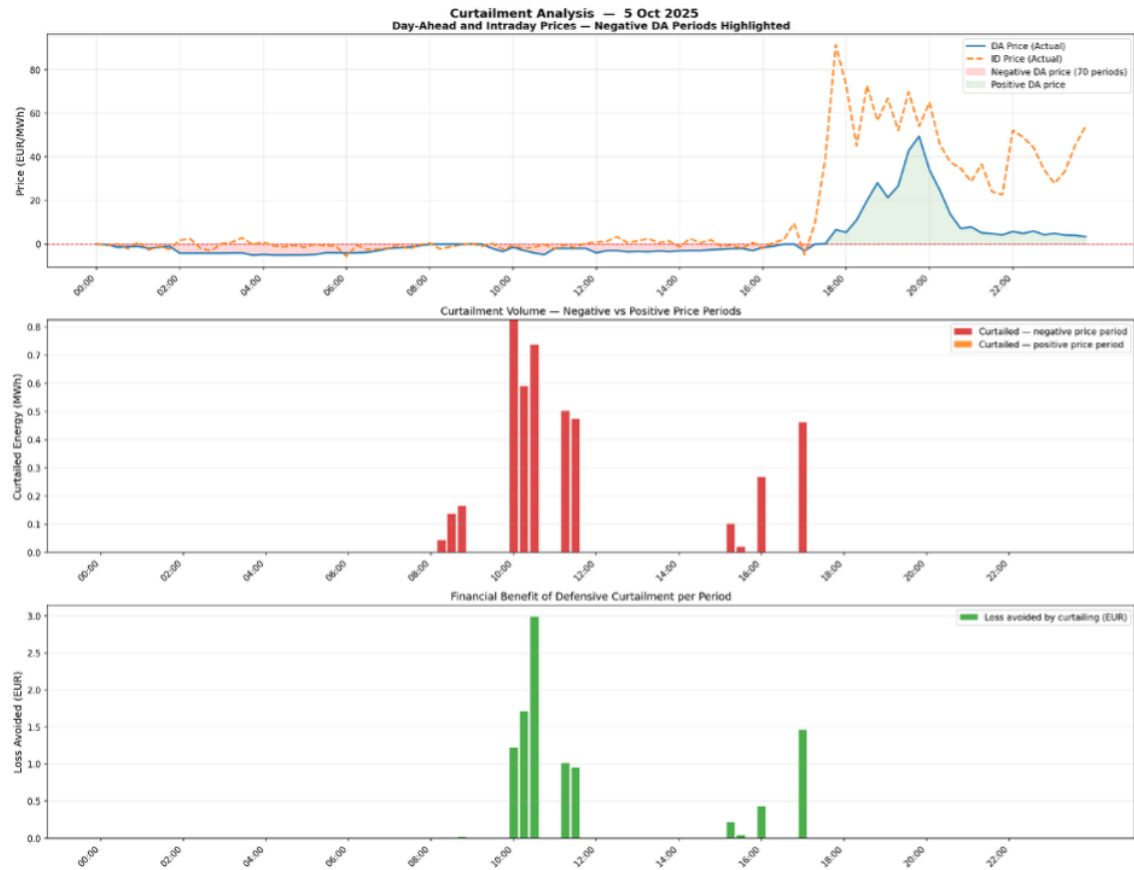


Figure D.7: Curtailment analysis.

D.9 aFRR Market Price Inputs and Threshold

Figure D.8 presents the aFRR market inputs and results through four subplots: BCap Neg capacity prices, EAM expected compensation prices, aFRR committed volumes, and generation against the 1 MW minimum bid threshold. The BCap price subplot shows clear variation across the six four-hour blocks of the day. Prices ranged from €8.00/MW in the late evening block (20:00–24:00) to €60.72/MW in the midday block (12:00–16:00), with the overnight blocks priced at €16.47/MW and €16.01/MW, the morning block at €34.99/MW, and the afternoon block at €24.95/MW. The actual mean BCap price across the day was €26.86/MW, slightly below the Forecast 2 estimate of €28.86/MW used at Stage 1. The EAM subplot shows the expected activation compensation across periods, averaging €10.53/MWh actual against the Forecast 2 estimate of €9.42/MWh, the EAM forecast slightly underestimated actual activation compensation on this day.

The aFRR volume subplot confirms 1.24 MW committed during the daylight four-hour blocks across all three stages. The optimizer concentrated commitment in the higher-priced midday and morning blocks where generation was sufficient to honour the obligation and the BCap reward was greatest. The minimum bid threshold subplot shows the 1 MW threshold line sitting below the generation profile during peak hours, confirming the asset meets the participation requirement throughout the daylight generation window.

The BCap capacity payment of €1,200.94 dominates the revenue picture, confirming that for a solar asset of this size the value of aFRR participation lies primarily in the capacity reservation payment rather than the energy activation compensation of €44.86. Together these form the offensive curtailment revenue of €1,245.81 (KPI 3), which at 98.0% of total revenue represents the transformative financial impact of aFRR market access for assets above the 1 MW minimum bid threshold.

D.10 Imbalance Analysis

Figure D.9 compares the imbalance outcomes of the two models across four subplots: benchmark positive imbalances, benchmark negative imbalances, optimized model imbalances, and settlement per period. A positive imbalance occurs when actual generation exceeds the total committed position across all markets — DA, ID, and aFRR EAM combined. A negative imbalance is the reverse — actual generation falls short of the total committed position. It is important to note that over-delivery can always be resolved through curtailment rather than creating a positive imbalance obligation, while under-delivery that cannot be curtailed represents a true imbalance obligation settled at the reBAP price.

Metric	Value
aFRR capacity committed	1.24 MW per active period
BCap Neg price overnight block (00:00–04:00)	€16.47/MW
BCap Neg price early morning block (04:00–08:00)	€16.01/MW
BCap Neg price morning block (08:00–12:00)	€34.99/MW
BCap Neg price midday block (12:00–16:00)	€60.72/MW
BCap Neg price afternoon block (16:00–20:00)	€24.95/MW
BCap Neg price late evening block (20:00–24:00)	€8.00/MW
BCap Neg mean (Forecast 2)	€28.86/MW
BCap Neg mean (Actual)	€26.86/MW
EAM mean (Forecast 2)	€9.42/MWh
EAM mean (Actual)	€10.53/MWh
BCap revenue	€1,200.94
EAM revenue	€44.86
Total offensive curtailment revenue (KPI 3)	€1,245.81

Table D.5: aFRR market participation and revenue.



Figure D.8: aFRR market price inputs and threshold.

The Benchmark Model incurred imbalances in 44 out of 96 periods — 4.38 MWh of over-delivery and 3.64 MWh of under-delivery relative to its DA commitment — arising from the mismatch between its DA commitment based on the generation forecast of 18.78 MWh and actual generation of 19.52 MWh. Since the benchmark commits only to the DA market with no ID correction, all deviations between actual generation and the DA forecast flow directly to imbalance settlement. The net imbalance settlement of €23.79 reflects the specific reBAP conditions on this day — the imbalance price was negative throughout 5 October 2025, meaning the system was oversupplied. Under negative reBAP, positive imbalances (over-delivery) incur a cost while negative imbalances (under-delivery) earn a gain because delivering less than committed is beneficial to an oversupplied system. On this day the negative imbalance gains outweighed the positive imbalance costs producing a net positive settlement. This outcome is entirely dependent on the sign of the reBAP and cannot be relied upon as a consistent strategy — on days with positive reBAP the benchmark would face severe losses from both imbalance types. A real trader could curtail surplus generation to eliminate positive imbalances — on this specific day this would produce a small improvement since positive imbalances were a cost — but the absence of ID hedging would still leave the benchmark exposed to large negative imbalances in under-delivery periods.

The Optimized Model incurred negative imbalances in 6 periods (6.2%), settling at

€-8.64. The optimized model’s imbalance reflects the full position — the difference between actual generation and the combined committed position across DA, ID, and aFRR EAM markets. The model correctly avoided all positive imbalances through curtailment — any period where actual generation would have exceeded the committed position was handled by withholding generation rather than over-delivering. The six negative imbalances are a structural constraint rather than a model flaw. They arise because in those specific periods the aFRR EAM dispatch obligation and the ID selling commitment together exceeded actual generation — a situation where curtailment cannot help since the problem is insufficient generation rather than excess. As discussed in Sections 5.4.1 and 5.4.3, imbalances are left as a decision variable in Stage 3 and penalised at €100,000/MWh to effectively prohibit deliberate imbalances. All 6 periods had negative DA prices. Of the 90 periods with no imbalance (93.8%), approximately 50 were overnight periods where solar generation was zero and no imbalance was possible regardless of market activity. Of the remaining active generation periods, the ID market successfully resolved forecast deviations in all but 6 cases, demonstrating effective hedging performance during the daylight hours when it was genuinely needed.

Table D.7 quantifies the role of the ID market in avoiding imbalances. Across 31 selling periods and 1 buying period, the ID market traded 14.58 MWh at an average price of €2.23/MWh, generating €32.49 in revenue. Without ID hedging, the DA commitment would have resulted in mismatches against actual generation — an average mismatch of 0.14 MWh per period, with a maximum over-commitment of 0.47 MWh and a maximum under-commitment of 1.08 MWh in a single period. Over-commitment mismatches can be resolved through curtailment rather than imbalance — only under-delivery mismatches that cannot be curtailed create a true imbalance obligation. The ID market addressed these under-delivery risks by buying back 0.14 MWh in 1 period, while the remaining generation surplus above DA commitments was sold through 31 ID selling periods, generating €32.49 in revenue at an average price of €2.23/MWh. This confirms that the ID hedging mechanism is not merely a revenue optimisation tool but an essential risk management instrument for any asset committing to both the DA and aFRR markets simultaneously. This result carries a direct implication for TokWise’s clients considering aFRR market entry. Participation in the aFRR capacity market introduces a binding obligation that, when combined with ID commitments, can create under-delivery situations that neither curtailment nor ID correction can resolve. As discussed in Section 5.2.1, this can only be mitigated through more accurate generation forecasting, more conservative reservation margins below the 67% applied here, or co-located BESS to guarantee the committed capacity can be delivered regardless of generation shortfalls.

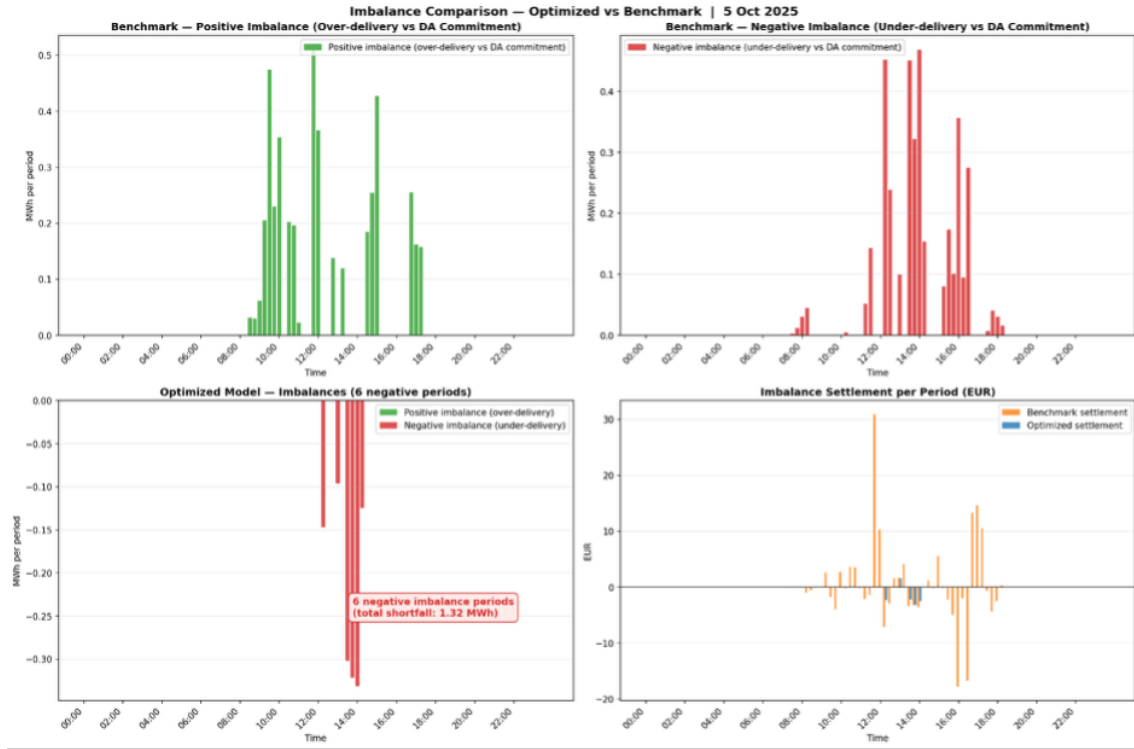


Figure D.9: Imbalance comparison — optimized vs benchmark.

Metric	Optimized Model	Benchmark Model
Periods with positive imbalance	0 (0.0%)	—
Periods with negative imbalance	6 (6.2%)	—
Periods with no imbalance	90 (93.8%)	—
Positive imbalance volume	0.00 MWh	4.38 MWh
Negative imbalance volume	1.32 MWh	3.64 MWh
Imbalance settlement	€-8.64	€23.79

Table D.6: Imbalance analysis and comparison statistics.

Metric	Value
Average DA vs actual mismatch	0.55 MW
Max positive mismatch (over-commitment)	1.87 MW
Max negative mismatch (under-commitment)	4.33 MW
Periods where ID market buys power	1
Periods where ID market sells power	31
Total sold via ID	14.58 MWh
Revenue from ID selling	€32.49
Avg price received	€2.23/MWh
Cost if same volume were imbalanced at €100,000/MWh	€1,405,562.50
Savings from using ID market instead	€1,405,595.17

Table D.7: Imbalance avoidance mechanism analysis.

Appendix E

Generation Forecast vs Actual — Error Analysis

Figure E.1 presents the generation forecast accuracy through three subplots: the forecast versus actual time series with shading indicating over- and under-forecast regions, the absolute forecast error per period, and the percentage forecast error per period. The forecast slightly underestimated actual generation throughout the daylight hours, with a total error of -0.73 MWh (-3.9%). The maximum over-forecast was 1.87 MW and the maximum under-forecast was -2.04 MW, reflecting the inherent uncertainty in solar generation forecasting at this asset scale. The forecast was essentially unbiased at an average error of 0.03 MW, indicating no systematic over- or under-prediction tendency on this day. However, the standard deviation of 0.64 MW confirms that while the forecast was accurate on average, individual period deviations were large enough in some instances to prevent full ID correction, directly contributing to the 6 negative imbalance periods observed at Stage 3.

The MAPE of 22.9% appears high but is largely a consequence of the many periods near zero where a small absolute error produces a large percentage figure. The absolute error metrics are more meaningful here and confirm acceptable forecast performance. The slight under-forecast of 0.73 MWh meant actual generation marginally exceeded the forecast, contributing a small amount of additional revenue compared to what a perfectly accurate forecast would have earned.

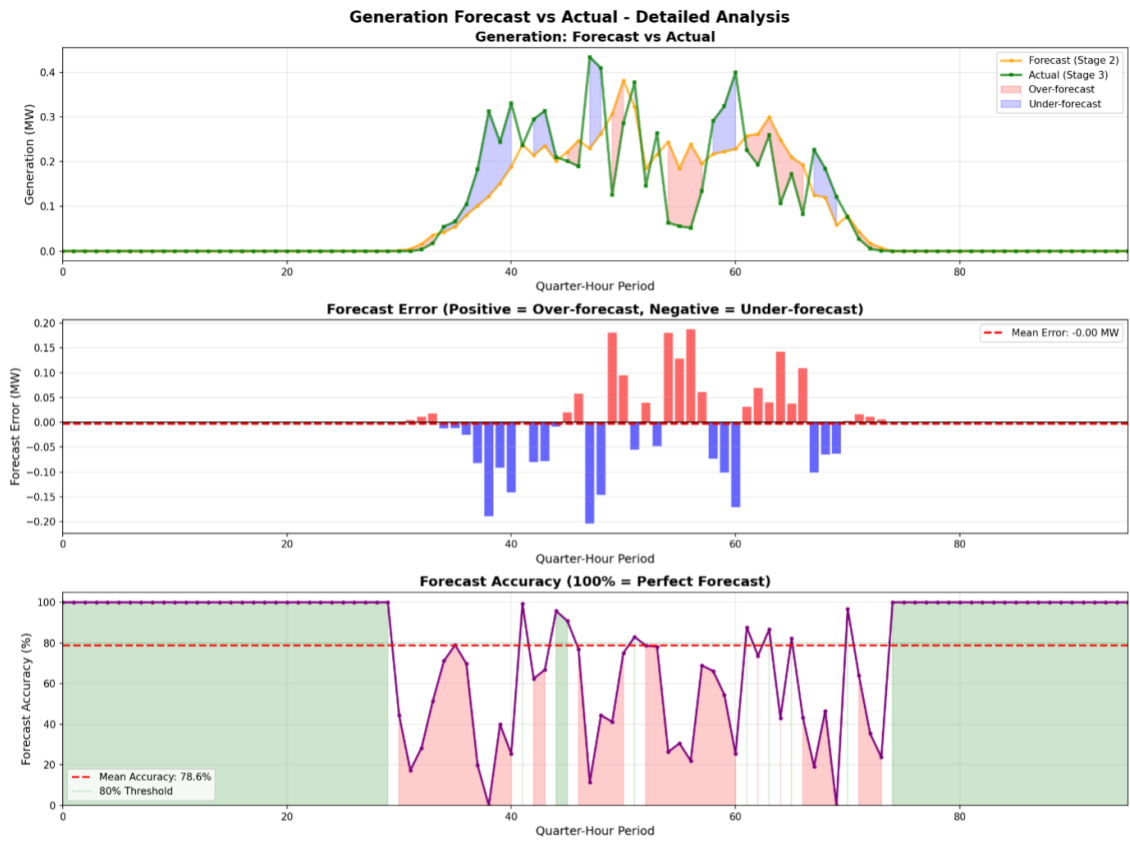


Figure E.1: Generation forecast vs actual - detailed analysis.

Metric	Forecast (Stage 2)	Actual (Stage 3)
Minimum	0.00 MW	0.00 MW
Maximum	3.81 MW	4.33 MW
Average	0.78 MW	0.81 MW
Total	18.78 MWh	19.52 MWh
Forecast error (total)	—	−0.73 MWh (−3.9%)
MAE	—	0.33 MW
RMSE	—	0.64 MW
MAPE	—	22.9%
Max over-forecast	—	1.87 MW
Max under-forecast	—	−2.04 MW
Accurate periods	—	58 / 96 (60.4%)
Forecast bias	—	−0.03 MW (unbiased)

Table E.1: Generation forecast error statistics.

Appendix F

Day-Ahead and Intraday Market Prices

Figure F.1 presents the actual and forecast DA and ID prices across all 96 periods. Two sets of statistics are reported in Table F.1 — full day across all 96 periods and filtered to the 42 periods where the asset had active solar generation. The distinction is important because the full-day statistics are significantly distorted by the late evening price spike to €49.41/MWh which occurred after solar generation had already ceased, and by the overnight negative prices which had no impact on dispatch decisions since generation was zero.

The DA price subplot shows two defining features of this day: sustained negative prices across the overnight, morning, and early afternoon hours, and a sharp evening spike reaching €49.41/MWh between 19:30 and 20:00. The full-day DA mean of €1.73/MWh conceals the true price environment the optimizer faced — during the 42 active generation periods, the DA mean was €-1.40/MWh with 38 out of 42 periods (90.5%) negative. This near-universal negative DA price environment during generation hours is the primary driver of the model’s core decision to minimise DA commitment and route the majority of generation through the ID market instead.

The DA forecast, constructed using the previous day’s prices as a naive persistence method, correctly captured this negative price environment, predicting 79 negative periods against 70 actual, but entirely missed the evening spike since October 4th’s maximum was only €4.02/MWh. Despite this magnitude error of €-2.02/MWh across the full day, the directional signal was sufficient to drive the correct dispatch decisions at both Stage 1 and Stage 2, the forecast correctly indicated that DA prices during generation hours would be predominantly negative, which is the only signal the optimizer needed to prefer ID over DA.

The ID price subplot shows better forecast accuracy with a mean error of €0.52/MWh. The ID forecast closely tracked actual Intraday prices throughout the day, confirming that the Intraday price forecast is reliable enough to support Intraday commitment decisions in Stage 3. During the 42 active generation periods, the actual Intraday price averaged €6.22/MWh with a median of €0.46/MWh, positive on average despite 18 out of 42 periods (42.9%) having negative ID prices, reflecting the high volatility of the ID market on this day. The ID maximum of €91.36/MWh during a generation period confirms the ID market’s potential to deliver significantly above-average returns in specific periods, which the optimizer exploited by concentrating ID sales in the highest-priced periods. The higher full-day ID mean of €12.93/MWh reflects the elevated evening prices after generation had ceased and does not represent the price conditions the optimizer faced during active dispatch decisions. The positive generation-period ID mean of €6.22/MWh against the generation-period DA mean of €-1.40/MWh, a spread of €7.62/MWh, confirms ID as the clearly

superior market for energy sales on average across the generation window on this day.



Figure F.1: Day ahead and intraday market prices.

Metric	DA Actual	DA Forecast	ID Actual	ID Forecast
Full day (all 96 periods)				
Mean [€/MWh]	1.73	-0.29	12.93	13.45
Min [€/MWh]	-5.09	-5.02	-5.66	-5.67
Max [€/MWh]	49.41	4.02	91.36	94.19
Forecast error (mean) [€/MWh]	-	-2.02	-	0.52
Negative periods	70/96 (72.9%)	79/96 (82.3%)	42/96 (43.8%)	42/96 (43.8%)
During generation periods only (42 active periods)				
Mean [€/MWh]	-1.40	-	6.22	-
Median [€/MWh]	-2.02	-	0.46	-
Min [€/MWh]	-4.88	-	-4.91	-
Max [€/MWh]	11.00	-	91.36	-
Negative periods	38/42 (90.5%)		18/42 (42.9%)	

Table F.1: DA and ID market price statistics.

Bibliography

- [1] P. Jowett. “Germany, France, Netherlands curtail 3.9 TWh of renewables in 2025,” pv magazine International, Accessed: Apr. 17, 2026. [Online]. Available: <https://www.pv-magazine.com/2026/02/03/germany-france-netherlands-curtail-3-9-twh-of-renewables-in-2025/>.
- [2] S. Enkhardt. “PV curtailment jumps 97% in Germany in 2024,” pv magazine International, Accessed: Apr. 17, 2026. [Online]. Available: <https://www.pv-magazine.com/2025/04/03/pv-curtailment-jumps-97-in-germany-in-2024/>.
- [3] Delfos. “Curtailment by Any Name: The €7.2B Problem Draining Renewable Generators Worldwide,” Accessed: Apr. 17, 2026. [Online]. Available: <https://www.delfos.energy/blog-posts/curtailment-has-many-names>.
- [4] “Germany - Countries & Regions,” IEA, Accessed: Apr. 21, 2026. [Online]. Available: <https://www.iea.org/countries/germany>.
- [5] B. Petrovich. “European Electricity Review 2026,” Ember, Accessed: Apr. 17, 2026. [Online]. Available: <https://ember-energy.org/latest-insights/european-electricity-review-2026>.
- [6] T. Oliveira. “Types of Curtailments: Economic and Technical,” Synertics, Accessed: Apr. 18, 2026. [Online]. Available: <https://synertics.io/blog/139/types-of-curtailments-economic-and-technical>.
- [7] H. K. Jacobsen and S. T. Schröder, “Curtailment of renewable generation: Economic optimality and incentives,” *Energy Policy*, Special Section: Fuel Poverty Comes of Age: Commemorating 21 Years of Research and Policy, vol. 49, pp. 663–675, Oct. 1, 2012, ISSN: 0301-4215. DOI: 10.1016/j.enpol.2012.07.004. Accessed: Apr. 18, 2026. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S030142151200585X>.
- [8] “AI-powered SaaS for renewables,” Accessed: Apr. 18, 2026. [Online]. Available: <https://www.tokwise.com/>.
- [9] “Description of concepts for balancing and the balancing markets in Germany,” Jan. 2025.
- [10] “Electricity market in transition - this is how electricity market 2.0 works,” SMARD, Accessed: Apr. 18, 2026. [Online]. Available: <https://www.smard.de/page/en/topic-article/205470/5972/electricity-market-in-transition-this-is-how-electricity-market-2-0-works>.
- [11] C. O’Connor, M. Bahloul, S. Prestwich, and A. Visentin, “A Review of Electricity Price Forecasting Models in the Day-Ahead, Intra-Day, and Balancing Markets,” *Energies*, vol. 18, no. 12, p. 3097, Jan. 2025, ISSN: 1996-1073. DOI: 10.3390/en18123097. Accessed: Apr. 18, 2026. [Online]. Available: <https://www.mdpi.com/1996-1073/18/12/3097>.

- [12] K.-M. Ehrhart and F. Ocker, “Allocation, Costs and Prices in the Electricity Wholesale Market and Balancing Power Market – An Integrated Approach,” [Online]. Available: <https://www.zew.de/fileadmin/FTP/veranstaltungen/Papers/Ocker.pdf>.
- [13] G. Zachmann, L. Hirth, C. Heussaff, I. Schlecht, J. Mühlenpfordt, and A. Eicke, “The design of the European electricity market - Current proposals and ways ahead,” Sep. 2023.
- [14] “Nord Pool — Data Portal,” Accessed: Apr. 21, 2026. [Online]. Available: <https://data.nordpoolgroup.com/auction/day-ahead/prices?deliveryDate=latest¤cy=EUR&aggregation=DeliveryPeriod&deliveryAreas=AT,GER>.
- [15] “Product Specifications - Western Central Europe Market Area,” Nord Pool AS, Sep. 30, 2025. [Online]. Available: <https://www.nordpoolgroup.com/4a813e/globalassets/download-center/rules-and-regulations/product-specifications-wce-september-30.pdf>.
- [16] “Trading Products — EPEX SPOT,” Accessed: Apr. 21, 2026. [Online]. Available: <https://www.epexspot.com/en/tradingproducts>.
- [17] “Home — EPEX SPOT,” Accessed: Apr. 21, 2026. [Online]. Available: <https://www.epexspot.com/en>.
- [18] K. Ganz and T. Kern. “An overview of the balancing services in Germany,” FfE, Accessed: Apr. 18, 2026. [Online]. Available: <https://www.ffe.de/en/publications/die-regelreservemaerkte-in-deutschland-im-ueberblick/>.
- [19] “Tendering and bidding process,” Accessed: Apr. 18, 2026. [Online]. Available: <https://www.regelleistung.net/en-us/Become-a-balancing-service-provider/Tendering-and-bidding-process>.
- [20] “Renewable Energy Sources Act (EEG) explained,” Accessed: Apr. 18, 2026. [Online]. Available: <https://montel.energy/resources/blog/what-is-the-renewable-energy-sources-act-eeeg>.
- [21] “Germany’s EEG subsidies,” Accessed: Apr. 18, 2026. [Online]. Available: <https://montel.energy/resources/blog/future-of-german-eeeg-subsidies-is-the-system-still-sustainable>.
- [22] “Negative wholesale prices,” Accessed: Apr. 18, 2026. [Online]. Available: <https://www.smard.de/page/en/wiki-article/5884/105426/negative-wholesale-prices?>.
- [23] “EEG levy 2021,” Accessed: Apr. 18, 2026. [Online]. Available: <https://www.netztransparenz.de/en/Renewable-energies-and-levies/EEG/EEG-financing/EEG-levies/EEG-levy-2021>.
- [24] “What types of balancing services are there?” Accessed: Apr. 18, 2026. [Online]. Available: <https://www.regelleistung.net/en-us/Basics-of-balancing-services/What-types-of-balancing-services-are-there>.

- [25] “Uniform imbalance price (reBAP),” Accessed: Apr. 18, 2026. [Online]. Available: <https://www.netztransparenz.de/en/Balancing-Capacity/Imbalance-price/Uniform-imbalance-price-reBAP>.
- [26] E. Kraft, M. Russo, D. Keles, and V. Bertsch, “Stochastic optimization of trading strategies in sequential electricity markets,” *European Journal of Operational Research*, vol. 308, no. 1, pp. 400–421, Jul. 1, 2023, ISSN: 0377-2217. DOI: 10.1016/j.ejor.2022.10.040. Accessed: Apr. 19, 2026. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0377221722008505>.
- [27] “Capacity reserve,” Accessed: Apr. 19, 2026. [Online]. Available: <https://www.netztransparenz.de/en/Ancillary-Services/System-operations/Capacity-reserve>.
- [28] “Präqualifizierte Leistungen in Deutschland,” Jan. 2026.
- [29] “The European power market: FCR, aFRR, mFRR & wholesale,” Accessed: Apr. 19, 2026. [Online]. Available: <https://www.enspired-trading.com/blog/the-european-power-market>.
- [30] B. Willis. “German PV plant becomes first prequalified to provide secondary control power,” PV Tech, Accessed: Apr. 19, 2026. [Online]. Available: <https://www.pv-tech.org/german-pv-plant-becomes-first-prequalified-to-provide-secondary-control-power/>.
- [31] “Negative spot market price - overview tables,” Accessed: Apr. 18, 2026. [Online]. Available: <https://www.netztransparenz.de/en/Renewable-energies-and-levies/EEG/Transparency-requirements/Market-premium/Negative-spot-market-price-overview-tables>.
- [32] “EG-Förderung und -Fördersätze,” Accessed: Apr. 19, 2026. [Online]. Available: <https://www.bundesnetzagentur.de/DE/Fachthemen/ElektrizitaetundGas/ErneuerbareEnergien/EEG.Foerderung/start.html>.
- [33] “Market value overview,” Accessed: Apr. 18, 2026. [Online]. Available: <https://www.netztransparenz.de/en/Renewable-energies-and-levies/EEG/Transparency-requirements/Market-premium/Market-value-overview>.
- [34] “Datacenter FCR/aFRR/mFRR/ABLA,” Accessed: Apr. 19, 2026. [Online]. Available: https://www.regelleistung.net/apps/datacenter/tenders/?productTypes=PRL%20CSRL%20MRL%20CABLA&markets=BALANCING_CAPACITY%20BALANCING_ENERGY&date=2024-11-13.
- [35] “PICASSO (Platform for the International Coordination of Automated Frequency Restoration and Stable System Operation),” Portail Services RTE, Accessed: Apr. 19, 2026. [Online]. Available: <https://www.services-rte.com/en/learn-more-about-our-services/picasso.html>.

- [36] M. Loesch, J. Rominger, S. Nainappagari, and H. Schmeck, “Optimizing Bidding Strategies for the German Secondary Control Reserve Market: The Impact of Energy Prices,” Jun. 1, 2018, pp. 1–5. DOI: 10.1109/EEM.2018.8469993.
- [37] C. Gärtner, E. Röger, and T. Heller. “An Optimal Bidding Model to Market Flexibility on the Balancing Electricity Markets,” Accessed: Apr. 19, 2026. [Online]. Available: <https://www.energy-proceedings.org/?p=10292>, pre-published.
- [38] J. Siemer. “German solar plant cleared to provide secondary control reserve,” pv magazine International, Accessed: Apr. 19, 2026. [Online]. Available: <https://www.pv-magazine.com/2025/11/17/german-solar-plant-cleared-to-provide-secondary-control-reserve/>.
- [39] L. Roald, S. Misra, M. Chertkov, S. Backhaus, and G. Andersson. “Chance Constrained Optimal Power Flow with Curtailment and Reserves from Wind Power Plants.” arXiv: 1601.04321 [math], Accessed: Apr. 19, 2026. [Online]. Available: <http://arxiv.org/abs/1601.04321>, pre-published.
- [40] L. Rüter, “Battery energy storage systems in hybrid power plants,” Technical University of Denmark, Jul. 2025. [Online]. Available: https://backend.orbit.dtu.dk/ws/portalfiles/portal/406232214/03072025_MasterThesis_LeonRuter_s233045_BEESinHPP_SizingOptimization.pdf.
- [41] dexterenergy. “Market impact in short-term power trading: The Dutch case and lessons for Belgium,” Dexter Energy, Accessed: Apr. 19, 2026. [Online]. Available: <https://dexterenergy.ai/news/market-impact-in-short-term-power-trading/>.
- [42] “Balancing markets,” TenneT, Accessed: Apr. 19, 2026. [Online]. Available: <https://www.tennet.eu/nl-en/markets/dutch-market/balancing-markets>.
- [43] “Electricity Mid-Year Update 2025 – Analysis,” IEA, Accessed: Apr. 19, 2026. [Online]. Available: <https://www.iea.org/reports/electricity-mid-year-update-2025>.
- [44] M. Brinkhaus. “Impact of the German EEG-law’s 1-hour rule,” Accessed: Apr. 19, 2026. [Online]. Available: <https://montel.energy/resources/blog/51-eeeg-effects-of-early-implementation-of-the-1-hour-rule>.
- [45] J. L. “Europe’s Power Paradox: Why Electricity Prices Went Below Zero in 2025,” Carbon Credits, Accessed: Apr. 19, 2026. [Online]. Available: https://carboncredits.com/europes-power-paradox-why-electricity-prices-went-below-zero-in-2025/?utm_source=chatgpt.com.
- [46] M. Pavlík, F. Kurimský, and K. Ševc, “Renewable Energy and Price Stability: An Analysis of Volatility and Market Shifts in the European Electricity Sector (2015–2025),” *Applied Sciences*, vol. 15, no. 12, p. 6397, Jan. 2025, ISSN: 2076-3417. DOI: 10.3390/app15126397. Accessed: Apr. 19, 2026. [Online]. Available: <https://www.mdpi.com/2076-3417/15/12/6397>.

- [47] “Growth in renewable energy in 2025,” Accessed: Apr. 19, 2026. [Online]. Available: https://www.bundesnetzagentur.de/SharedDocs/Pressemitteilungen/EN/2026/20260108_EEG.html.
- [48] “Electricity Storage Valuation Framework 2020,” IRENA, Mar. 5, 2020. Accessed: Apr. 19, 2026. [Online]. Available: <https://www.irena.org/publications/2020/Mar/Electricity-Storage-Valuation-Framework-2020>.
- [49] “Renewable Energy Market Update - June 2023,” IEA, 2023. [Online]. Available: https://iea.blob.core.windows.net/assets/63c14514-6833-4cd8-ac53-f9918c2e4cd9/RenewableEnergyMarketUpdate_June2023.pdf.
- [50] N. Nolzen, A. Ganter, N. Baumgärtner, L. Leenders, and A. Bardow. “Where to Market Flexibility? Optimal Participation of Industrial Energy Systems in Balancing-Power, Day-Ahead, and Continuous Intraday Electricity Markets.” arXiv: 2212.12507 [math], Accessed: Apr. 19, 2026. [Online]. Available: <http://arxiv.org/abs/2212.12507>, pre-published.